

1 Topological Spaces

Definition 1.1: Topology

Let X be any set. A topology T on X is a collection of subsets of X , which satisfies the following axioms.

- $\emptyset$ and X are both in T
- Closed under arbitrary unions
- Closed under finite intersections

The sets in T are called open sets.

A set X equipped w/ a topology is called a topological space.

Example:

text

- X is any set, $T = P(X)$. (discrete topology)
- X is any set, $T = \{\emptyset, X\}$. (indiscrete topology)
- $X = \{a, b, c\}$, $T = \{\emptyset, X, \{a, b\}\}$
- $X = \mathbb{R}$, $T = \{\emptyset, X\} \cup \{U \subset \mathbb{R} \mid \text{for all } x \in U, \exists \epsilon > 0 \text{ s.t. } (x - \epsilon, x + \epsilon) \subset U\}$
- X is any set, let T be the collection of subsets $U \subseteq X$ s.t. either $U = \emptyset$, or $X \setminus U$ is finite.

Proposition 1.2

This is a topology.

Proof.

- $\emptyset, \mathbb{R}$ are both open
- Suppose $\{U_j\}_{j \in J}$ is a collection of open sets in $\mathbb{R}$
Let $x \in \bigcup_{j \in J} U_j$. Then $x \in U_j$ for some $j \in J$.
 U_j is open thus there is an open interval in U_j containing x . So it is in the arbitrary union
- Let U, V be open subsets of $\mathbb{R}$

Let $x \in U \cap V$. Because U, V open there are intervals $(a, b) \subset U, (c, d) \subset V$ s.t. $x \in (a, b) \cap (c, d)$
 $(a, b) \cap (c, d) = (\max(a, c), \min(b, d)) \subset U \cap V$. Thus, x is in this interval thus $U \cap V$ is open.

Proposition 1.3

The "cofinite" topology is a topology.

Proof.

- The empty set is open, and $X \setminus X = \emptyset$ is finite, so $X \in T$
- Consider sets $\{U_j\}_{j \in J}$ s.t. $U_j \in T$
 First: consider the case for some j' , $X \setminus U_{j'}$ is finite.
 $X \setminus \bigcup_{j \in J} U_j = \bigcap_{j \in J} X \setminus U_j \subset X \setminus U_{j'}$. So the arbitrary union is finite.
 Second: If the first case does not occur, then $U_j = \emptyset$ for all $j \in J$, so the union is the empty set.
- Let $U, V \in T$. If either is the empty set then $U \cap V = \emptyset \in T$, so assume that both are not empty. Then consider $X \setminus (U \cap V) = (X \setminus U) \cup (X \setminus V)$, so it is a union of two finite sets which is finite thus $U \cap V \in T$

Definition 1.4: Closed Set

If X is a set with topology T , then the complement of an open set in X is called a closed set.

Example

If T is cofinite topology, a set in $A \subseteq X$ is closed $\iff$ either finite or $A = X$

IMPORTANT: Some sets are neither open nor closed. Some sets are both closed and open.

2 Basis for a topology

Definition 2.1: Basis

Let X be a set. A collection β of subsets of X is a basis for a topology if:

- every $x \in X$ is in some $B \in \beta$
- If $x \in B_1 \cap B_2$ for $B_1, B_2 \in \beta$
then: $\exists B_3 \in \beta$ s.t. $x \in B_3 \subseteq B_1 \cap B_2$

The topology generated by β is the collection T of subsets defined by:

$$U \in T \iff \text{for every } x \in U, \exists B \in \beta \text{ s.t. } x \in B \subseteq U$$

Example

$X = \mathbb{R}$, β = set of open intervals in $\mathbb{R}$

$X = \mathbb{R}^n$, β = set of open balls $B(x, r) = \{y \in \mathbb{R}^n \mid d(x, y) < r\}$

Check that the second example is actually a basis for a topology

- every $x \in \mathbb{R}^n$ lies in some open ball. $x \in B(x, 1)$
- Assume $x \in B_1 \cap B_2 = B(y_1, r_1) \cap B(y_2, r_2)$
Set $r_3 = \min(r_1 - d(x, y_1), r_2 - d(x, y_2))$
Consider the ball $B_3 = B(x, r_3)$. Let $z \in B_3$.
 $d(z, y_1) \leq d(x, y_1) + d(z, x) < d(x, y_1) + r_3 \leq d(x, y_1) + r_1 - d(x, y_1) = r_1$
So $z \in B(y_1, r_1) = B_1$ do the same proof for B_2 thus $z \in B_1 \cap B_2$ thus $B_3 \subseteq B_1 \cap B_2$

Proposition 2.2

This is actually a topology on X .

Proof.

- $\emptyset \in T$ vacuously, and $X \in T$ comes from first basis axiom.
- Suppose $\{u_j\}_{j \in J}$ of sets in T .
Let $x \in \bigcup_{j \in J} u_j$. Thus $x \in u_{j'}$ for some $j' \in J$, so $\exists B_{j'} \in \beta$ s.t. $x \in B_{j'} \subseteq u_{j'}$ thus
 $x \in B_{j'} \subseteq \bigcup_{j \in J} u_j$. This shows $\bigcup_{j \in J} u_j \in T$
- Suppose $x \in u_1 \cap u_2$ for $u_1, u_2 \in T$. Then $x \in B_1, B_2$ for sets $B_1, B_2 \in \beta$ s.t. $B_1 \subseteq u_1$

and $B_2 \subseteq u_2$ thus $x \in B_1 \cap B_2$. By the second axiom thus implies $\exists B_3 \in \beta$. Thus $x \in B_3 \subseteq B_1 \cap B_2 \subseteq u_1 \cap u_2 \in T$

Remark.

Every topology is a basis for itself.

Proposition 2.3

Let X be a set, let β be a basis for topology T . The sets in T are precisely the sets of the form

$$\bigcup_{j \in J} B_j$$

for $\{B_j\}_{j \in J}$ collection of sets in β

Proof.

- Every set of this form is in T :

$$x \in \bigcup_{j \in J} B_j, \text{ then } x \in B_{j'} \subset \bigcup_{j \in J} B_j$$

- Every set in T has the form:

$$\text{Let } U \in T. \text{ Then for every } x \in U, \text{ there is a } B_x \in \beta \text{ s.t. } B_x \subset U \implies U = \bigcup_{x \in U} B_x$$

Decomposition is NOT in general unique.

Basis sets are always open.

2.1 Comparing Topologies

Definition 2.4: Finer

If T_1, T_2 are topologies on the same set X , we say that T_1 is finer than T_2 if $T_1 \supseteq T_2$. Also, T_2 is coarser than T_1

Proposition 2.5

Let β, β' be bases for topologies T, T' on X . Then $T \subseteq T'$ (T' is finer than T) if and only if every $x \in X$, and every $B \in \beta$ containing x , $\exists$ some $B' \in \beta'$ s.t. $x \in B' \subset B$

Example

$X = \mathbb{R}^n$, let β be the set of open boxes in $\mathbb{R}^n$: sets of the form $(a_1, b_1) \times (a_2, b_2) \times \cdots \times (a_n, b_n)$

This is a basis. It generates the same standard topology on $\mathbb{R}^n$.

Example

Let $X = \mathbb{R}$. Consider collections of half-open intervals in $\mathbb{R}$: $[a, b), (a, b]$.

Claim: this is a basis for a topology.

This is not the same as the std. topology on $\mathbb{R}$

This is finer than the std. topology.

3 Examples of topological spaces

Example 3.1: Order Topology

Let X be a set with the order relation $<$. For $a < b$ in X , define intervals:

$$(a, b) = \{x \in X \mid a < x < b\}$$

$$[a, b) = \{x \in X \mid a \leq x < b\}$$

Similar for $(a, b], [a, b]$

Let β be the set of all intervals in X of the forms:

- (a, b) for $a < b$ in X .
- $[a_0, b)$ where a_0 is the smallest element of X .(if it exists).
- $(a, b_0]$ where b_0 is the largest element of X .(if it exists).

Proposition 3.2

This is a basis for a topology on X .

This is called the order topology on X .

Example

Standard topology on $\mathbb{R}$ is order topology, on $\mathbb{R}$

Example 3.3: Product Topology

Let X, Y be topological spaces.

Let β be a collection of sets:

$$\{U \times V \mid U \text{ open in } X, V \text{ open in } Y\}$$

If β_1 is a basis for topology on X , and β_2 is a basis for topology on Y , then

$$\{B \times C \mid B \in \beta_1, C \in \beta_2\}$$

is a basis for the product topology.

Proposition 3.4

This is a basis for a topology.

This is called the product topology on $X \times Y$

Example

The product topology is std topology on $\mathbb{R}$

This definition is valid for products of pairs of sets.

Lemma 3.5

Suppose X is a topological space. Let C be a collection of open subsets satisfying:

- for each open set $U \subseteq X$ and each $x \in U$, $\exists c \in C$ s.t. $x \in c \subseteq U$

Then, C is a basis for top. on X .

Proof.

Check C is a basis.

- Every $x \in X$ in some $c \in C$. X is open, so there exists c s.t. $x \in c \subseteq X$
- If $c_1, c_2 \in C$, then for any $x \in c_1 \cap c_2$, $\exists c_3 \in C$ s.t. $x \in c_3 \subseteq c_1 \cap c_2$.
Follows from the fact that c_1, c_2 are open which implies $c_1 \cap c_2$ is open so can find c_3

Check: C generates same topology T on X .

First, suppose U is open in T . By assumption for all $x \in U$, there exists $c \in C$ s.t. $x \in c \subseteq U$, so U is open in the topology generated by C .

If U is open in topology generated by C . Then U is a union of sets in C . But every $c \in C$ is open in T , so union U is also open in T . ■

Definition 3.6: Metric

Let X be a set. A function $d : X \times X \rightarrow \mathbb{R}$ is a metric if it satisfies:

- $d(x, y) = 0 \iff x = y$
- $d(x, y) = d(y, x) \quad \forall x, y \in X$
- $d(x, y) + d(y, z) \geq d(x, z) \quad \forall x, y, z \in X$

Example 3.7: Metric Topology

A set X equipped with a metric d is a metric space.

Let β be the set of all metric balls:

$$B_d(x, r) = \{y \in X \mid d(x, y) < r\} \quad \text{for } x \in X, r > 0$$

Proposition 3.8

This is a basis for a topology

Definition 3.9: Metrizable

A topological space X is called metrizable if $\exists$ metric d on X inducing topology on X .

Remark.

There could be more than one metric inducing the same topology.

Example 3.10: Subspace Topology

Let X be a set with topology $\mathcal{T}$. Let $Y \subseteq X$.

Thus subspace topology on Y is the collection of sets:

$$\{U \cap Y \mid U \in \mathcal{T}\}$$

Proposition 3.11: $\mathcal{T}$

is a topology on Y .

We call Y a subspace of topology X .

Example

Suppose (X, d) is a metric space. $Y \subseteq X$.

The function d is also a metric on Y . We have 2 topologies on Y :

- Subspace topology as a subspace of X with metric topology.
- Metric topology on Y

Proposition 3.12

These two topologies are the same.

Proof.

Let $\mathcal{T}_{\subseteq}$ is subspace topology, and $\mathcal{T}_d$ is metric topology on Y .

$\mathcal{T}_d$ has a basis $\beta_d = \{B_d(y, r) \cap Y \mid y \in Y, r > 0\}$

$\mathcal{T}_{\subseteq}$ has a basis $\beta_{\subseteq} = \{B_d(x, r) \cap Y \mid x \in X, r > 0\}$

Subspace Basis is just X 's basis but intersected with Y .

Easy to see that $\beta_d \subseteq \beta_{\subseteq}$ thus $\mathcal{T}_d \subseteq \mathcal{T}_{\subseteq}$

Let $y \in B \in \beta_{\subseteq}$, so $y \in B_d(x, r) \cap Y$ for some $x \in X, r > 0$

Using the triangle inequality, find some $r' > 0$ s.t.

$$B_d(y, r') \subseteq B_d(x, r)$$

Then

$$B_d(y, r') \cap Y \subseteq B_d(x, r) \cap Y$$

Example

Let X, Y be nonempty topological spaces. Equip $X \times Y$ with the product topology. For each $y \in Y$, the set $X \times \{y\}$ is a subspace of $X \times Y$, and $X \times \{y\}$ is in bijection with X . Two topologies on X :

- Original topology
- Subspace topology on $X \times \{y\}$ identified with X .

Proposition 3.13

These two topologies are the same.

Example

Let X be a set with an order $<$. Let $Y \subseteq X$. Then $<$ gives an order relation of Y . Two topologies on Y :

- Order topology on Y
- Subspace topology

Proposition 3.14

These do not have to agree!.

Proof.

Let $X = \mathbb{R} \times \mathbb{R}$, with dictionary order and $Y = [0, 1] \times [0, 1]$. Consider $\{1/2\} \times [0, 1]$:
This is open in the subspace topology since $\{1/2\} \times (-\epsilon, 1 + \epsilon)$ is an interval in X .
This is not open on the order topology since it has the endpoints. ■

4 Continuous Functions

Definition 4.1: Continuous Function

Let X, Y be topological spaces. A function $f : X \rightarrow Y$ is continuous if, for every open $U \subseteq Y$, the set $f^{-1}(U)$ is open in X .

Definition 4.2: Homeomorphism

A bijection $f : X \rightarrow Y$ is called a homeomorphism if it is continuous and $f^{-1} : Y \rightarrow X$ is also continuous.

Equivalently: If $f : X \rightarrow Y$ is a bijection, then it induces a bijection $P(X) \rightarrow P(Y)$. If X, Y are topological spaces, a bijection $f : X \rightarrow Y$ is a homeomorphism if it identifies topology on X with the topology on Y .

Example

Let $X = \{a, b\}$ and $Y = \{c, d\}$. Let $\mathcal{T}_X = \{\emptyset, X, \{a\}\}$. Let $\mathcal{T}_Y = \{\emptyset, Y\}$. Consider $f : X \rightarrow Y$ s.t. it is a bijection this is clearly continuous since $f^{-1}(Y) = (X)$. However for $f^{-1} : Y \rightarrow X$ this is not continuous.

Definition 4.3: Embedding

Let $f : X \rightarrow Y$, let $Z \subseteq Y$ be the subspace $f(X)$. If $f : X \rightarrow Z$ is a homeomorphism, then we say that f is a topological embedding.

This is not the same as a continuous injective map.

Example

Let $X = [0, 2\pi)$, $f : X \rightarrow \mathbb{R}^2$ given by $f(t) = (\cos(t), \sin(t))$.

Proposition 4.4

Let $f : X \rightarrow Y$ be a map between topological spaces X, Y .

$$f \text{ is continuous} \iff f^{-1}(B) \text{ is open for every } B \text{ in a basis for } Y.$$

Proof.

($\implies$) : immediate by the fact that basis sets are open.

($\impliedby$) : use fact that every open set in Y is a union of basis sets. ■

Definition 4.5: Neighborhood

Let X be a topological space, $x \in X$. A neighborhood of x is any open set $U \subset X$ containing x .

Definition 4.6: Continuous at a Point

A map $f : X \rightarrow Y$ between topological spaces is continuous at $x \in X$ if, for every neighborhood V of $f(x)$, there is a neighborhood U of x s.t. $f(U) \subseteq V$.

Proposition 4.7

Let $f : X \rightarrow Y$ be a map between topological spaces then the following are equivalent.

- (i) f is continuous
- (ii) f is continuous at every $x \in X$
- (iii) $f^{-1}(C)$ is closed for every closed $C \subseteq Y$

Proof.

(i) $\iff$ (iii) : Let $U \subseteq Y$ be open.

Thus, $X \setminus f^{-1}(U) = f^{-1}(Y) \setminus f^{-1}(U) = f^{-1}(Y \setminus U)$

Thus, $f^{-1}(U)$ is open $\iff f^{-1}(Y \setminus U)$ is closed.

(i) $\implies$ (ii) : Let $x \in X$, let V be a neighborhood of $f(x)$. Then $f^{-1}(V)$ is open, so it's a neighborhood of x and $f(f^{-1}(V)) \subseteq V$.

(ii) $\implies$ (i) : Let $V \subseteq Y$. For each $x \in f^{-1}(V)$, $f(x) \in V$, so there is a neighborhood U_x of x s.t. $f(U_x) \subseteq V$, meaning $U_x \subseteq f^{-1}(V)$.

Thus, $f^{-1}(V) = \bigcup_{x \in f^{-1}(V)} U_x$, so $f^{-1}(V)$ is open. ■

Definition 4.8: Continuous on Metric Spaces

Let X, Y be metric spaces. A function $f : X \rightarrow Y$ is continuous at $x \in X$ if $\forall \epsilon > 0$, $\exists \delta > 0$ s.t. whenever $d(x, y) < \delta$, then $d(f(x), f(y)) < \epsilon$.

Also the same as, for every ball $B(f(x), \epsilon)$, $\exists \delta$ s.t. $f(B(x, \delta)) \subseteq B(f(x), \epsilon)$.

Proposition 4.9

These definition of continuity are equivalent when X, Y are metric spaces.

Lemma 4.10

Let $f : X \rightarrow Y, g : Y \rightarrow Z$ be continuous between topological spaces. Then $g \circ f$ is continuous.

Proof.

Let $U \subseteq Z$ be open then $(g \circ f)^{-1}(U) = f^{-1}(g^{-1}(U))$. $g^{-1}(U)$ is open, since g is continuous then $f^{-1}(g^{-1}(U))$ is open, since f is continuous. ■

Proposition 4.11

Let X be a topological space, $A \subseteq X$ a subspace then the inclusion $f : A \hookrightarrow X$ is continuous.

Proof.

Let $U \subseteq X$ be open, thus $f^{-1}(U) = U \cap A$ is open in A by definition of subspace topology. ■

Proposition 4.12

Let $f : X \rightarrow Y$ be a map between topological spaces, suppose that $f(X) \subseteq Z$ for some subspace $Z \subseteq Y$. Then $f : X \rightarrow Z$ is continuous $\iff f : X \rightarrow Y$ is continuous.

Proposition 4.13

If $f : X \rightarrow Y$ is continuous, and A is a subspace of X then $f|_A : A \rightarrow Y$ is continuous.

Definition 4.14: Open Covering

Let X be a topological space. An open covering of X is a collection $\{U_\alpha\}_{\alpha \in J}$ of open sets s.t. $X = \bigcup_{\alpha \in J} U_\alpha$.

Proposition 4.15

Let $f : X \rightarrow Y$, let $\{U_\alpha\}_{\alpha \in J}$ be an open covering. Then f is continuous $\iff$ each restriction $f|_{U_\alpha}$ is continuous.

Proposition 4.16

Suppose X is a union of finitely many closed subsets $C_1, \dots, C_n$.
Then a map $f : X \rightarrow Y$ is continuous $\iff$ each restriction $f|_{C_i} : C_i \rightarrow Y$ is continuous.

Let $\mathbb{R}$ have std. top.

- Addition
- Multiplication
- Division (x/y) for $y \neq 0$

These are continuous.

Lemma 4.17

If $f_1 : X_1 \rightarrow Y_1, f_2 : X_2 \rightarrow Y_2$ are continuous maps of top spaces, then the map $X_1 \times X_2 \rightarrow Y_1 \times Y_2$ given by $(u, v) \mapsto (f_1(u), f_2(v))$ is continuous.

5 Closures of Sets in Topological Spaces

Definition 5.1: Closure of a set

Let X be a topological spaces, $A \subseteq X$. The closure of A in X , denoted $\bar{A}$, is the intersection of all closed subsets of X containing A . (The "smallest" closed subset containing A)

Definition 5.2: Interior of a set

The interior of A is the unions of all open subsets contained in A .

$$\text{Int}(A) = X \setminus \overline{X \setminus A}$$

Theorem 5.3

Let X be a top. space, let $A \subseteq X$. Then $x \in \bar{A} \iff$ every neighborhood of x intersects A .

Proof.

This is the same as $x \notin \bar{A} \iff$ there is some neighborhood of x which does not intersect $A \iff$ there exists a closed subset of X containing A , and not x .

This is equivalent to $x \notin \bar{A}$ since it is not in some closed set containing A . ■

Corollary 5.4

Suppose β is a basis for topology on X . Then $x \in \bar{A} \iff$ every $B \in \beta$ containing x intersects A .

Proof.

($\implies$) this is proved from the last theorem since basis sets are open.

($\impliedby$) Let U be a neighborhood of x . Then, there is some $B \in \beta$ s.t. $x \in B \subseteq U$. Know B intersects A , so U also intersects A . This means every neighborhood of x intersects A , so $x \in \bar{A}$ ■

Example: Let $X = \mathbb{R}^2$, let $A = B(0,1)$ then $\bar{A} = \bar{B}(0,1) = \{x \in \mathbb{R}^2 \mid d(0,x) \leq 1\}$

First, supposed $x \notin \bar{B}(0,1)$. Then $d(0,x) = r > 1$. Using the triangle inequality, can see that $B(x, r-1)$ does not intersect $B(0,1)$. Thus, $x \notin \bar{B}(0,1)$.

Suppose $x \in \bar{B}(0,1)$. Then $d(0,x) \leq 1$. Fix $\epsilon > 0$. Consider the point $y = (1-\epsilon)x$. Thus, $d(x,y) = d(x, (1-\epsilon)x) = d(x, x-\epsilon x) = d(0, -\epsilon x)$ is given by $\|\epsilon x\| = \epsilon \|x\| \leq \epsilon$

$y \in B(x, 2\epsilon)$. Also, $d(0, y) = d((1 - \epsilon)x, 0) = (1 - \epsilon)||x|| \leq 1 - \epsilon < 1$, so $y \in B(0, 1)$. Thus, every ball about x intersects $B(0, 1)$, so $x \in \overline{B(0, 1)}$ ■

Remark.

- $\bar{\bar{A}} = \bar{A}$
- A is closed $\iff A = \bar{A}$

Definition 5.5: Accumulation Point

Let A be a subset of a space X . An accumulation point of A is a point $x \in X$ s.t. every neighborhood of x intersects A in a point not equal to x .

Example

Every point in $\mathbb{R}$ is an accumulation point of $\mathbb{Q}$.

No points in $\mathbb{R}$ are accumulation points in $\mathbb{N}$.

If $A = \{1/n \mid n \in \mathbb{N}\}$, then 0 is the only accumulation point of A . ■

Theorem 5.6

Let X be a topological space, let $A \subseteq X$, and A' be the set of all accumulation points of A . Then $\bar{A} = A \cup A'$.

Proof.

Show: $A \cup A' \subseteq \bar{A}$

Know: $A \subseteq \bar{A}$. So let $x \in A'$

Then every neighborhood of x intersects A , so $x \in \bar{A}$.

Show: $\bar{A} \subseteq A \cup A'$. Let $x \in \bar{A}$. If $x \in A$, we are done. So, assume $x \notin A$. Know every neighborhood of x intersects A . Since $x \notin A$, this neighborhood must actually intersect $A \setminus \{x\} = A$ ■

Definition 5.7: Converges

Let x_n be a sequence in a space X . We say that x_n converges to $x \in X$ if, for all neighborhoods U of x , $\exists N \in \mathbb{N}$ s.t. $x_n \in U$ for all $n \geq N$.

Example

Let $X = \{a, b, c\}$. Let $\mathcal{T} = \{\emptyset, X, \{a, b\}\}$.

Consider the sequence $x_n = a$ converges to a , since it is in every neighborhood of a ,

but it also converges to b since x_n is in every neighborhood of b . It also converges to c since x_n is in every neighborhood of c . ■

Example

If X is a space with indiscrete topology then every sequence converges to every point in X . ■

Definition 5.8: Hausdorff Space

A topological space is called Hausdorff if it satisfies the following:

- For every pair of distinct points x_1 and x_2 , there are neighborhoods U_1 of x_1 and U_2 of x_2 s.t. $U_1 \cap U_2 = \emptyset$.

Theorem 5.9

Let x_n be a sequence in a Hausdorff space. Then x_n converges to at most one point.

Proof.

Suppose x_n converges to x . Let y be any point distinct from x .

WTS: x_n does not converge to y .

Fix neighborhoods U of x and V of y s.t. $U \cap V = \emptyset$. For all but finitely many n , $x_n \in U$. This means $x_n \in V$ for only finitely many n , so x_n cannot converge to y . ■

Theorem 5.10

Let X be a Hausdorff space. Then any singleton $\{x\}$ is a closed subset of X .

Proof.

WTS: $X \setminus \{x\}$ is open.

For each $y \in X \setminus \{x\}$, there is a neighborhood U_y of x and V_y of y s.t. $U_y \cap V_y = \emptyset$. In particular $V_y \subseteq X \setminus \{x\}$. Thus:

$$X \setminus \{x\} = \bigcup_{y \in X \setminus \{x\}} U_y$$

is open. ■

In that proof we really only needed: for every pair of distinct points $x, y \in X$, there exists a neighborhood U of x not containing y . (T_1 property)

Proposition 5.11

Every metrizable space X is Hausdorff.

Proof.

Let $x, y \in X$. Set $r = d(x, y)$, then by the triangle inequality, $B(x, r/2) \cap B(y, r/2) = \emptyset$. ■

Proposition 5.12

Let X be a top. space, let $A \subseteq X$. Suppose x_n is a sequence in A , and $x_n \rightarrow x$ in X . Then $x \in \bar{A}$

Proof.

Let U be a neighborhood of x . Then $x_n \in U$ for all but finitely many n , so $U \cap A \neq \emptyset \implies x \in \bar{A}$ ■

Proposition 5.13

Assume X is metrizable. Then, every $x \in \bar{A}$ is a limit of a sequence x_n for $x_n \in A$

Proof.

Let $x \in \bar{A}$. For every $n \in \mathbb{N}$, the ball $B(x, 1/n)$ intersects A . So, let $x_n \in A \cap B(x, 1/n)$. If U is any neighborhood of x , then U contains $B(x, \epsilon)$ for some $\epsilon > 0$. Then $x_n \in B(x, \epsilon)$ whenever $n > 1/\epsilon$. Thus, $x_n \rightarrow x$. ■

Theorem 5.14

Let $f : X \rightarrow Y$ be a map of topological spaces. Then
 f is continuous $\implies$ for any sequence x_n in X converging to a point $x \in X$ we have $f(x_n) \rightarrow f(x)$
 If X is metrizable $\iff$ holds.

6 Quotient Topology

Let X be a top. space, let $\sim$ be an equivalence relation on X . There is a surjective map $\rho : X \rightarrow X/\sim$ $x \mapsto [x]$

$X/\sim$ is the set of equivalence classes with respect to $\sim$.

Definition 6.1: Quotient Topology

The quotient topology on $X/\sim$ is the collection of sets $U \subseteq X/\sim$ s.t. $\rho^{-1}(U)$ is open in X .

This is the finest topology on $X/\sim$ s.t. $\rho : X \rightarrow X/\sim$ is continuous.

Definition 6.2: Quotient Map

In general, if X, Y are top. spaces, a surjective map $\rho : X \rightarrow Y$ is a quotient map if it satisfies:

$$U \subseteq Y \text{ open} \iff \rho^{-1}(U) \text{ open}$$

When this occurs, Y can be identified homeomorphically to a quotient space for X , under the relation $x_1 \sim x_2 \iff \rho(x_1) = \rho(x_2)$

If X, Y are top. spaces, can form disjoint union $X \sqcup Y$. Topology on $X \sqcup Y$ is the collection $U \sqcup V$ for $U \subseteq X, V \subseteq Y$ open.

Suppose X_1, X_2 spaces, $Y_1 \subseteq X_1, Y_2 \subseteq X_2$ subspaces, $\phi : Y_1 \rightarrow Y_2$ is a homeomorphism.

Can form space $X_1 \sqcup_{\phi} X_2 = X_1 \sqcup X_2 / \sim_{\phi}$

$\sim_{\phi}$ is relation given by: $x \sim_{\phi} x$ and $y \sim \phi(y)$ and $\phi(y) \sim y$ for every $y \in Y_1$

Theorem 6.3

Suppose $f : X \rightarrow Y$ is map, and $\sim$ is an equivalence relation. There is a map $X/\sim \rightarrow Y$ is $f(x) = f(y)$ whenever $x \sim y$

If $X/\sim$ has the quotient topology, then f is continuous $\iff f' : X/\sim \rightarrow Y$ is continuous.

7 Product Spaces (revisited)

Let $\{X_\alpha\}_{\alpha \in J}$ be a collection of sets. Can form Cartesian product $\prod_{\alpha \in J} X_\alpha$.

If $X_\alpha = X$, denoted $X^J \iff \text{map: } J \rightarrow X$

If each X_α top spaces, first guess for topology on $\prod_{\alpha \in J} X_\alpha$

Definition 7.1: Box topology

Topology generated by the basis $\prod_{\alpha \in J} U_\alpha$ s.t. each U_α open in X_α .

We have natural maps associated to product:

$$\pi_\alpha : \prod_{\alpha \in J} X_\alpha \rightarrow X_\alpha$$

for each $\alpha \in J$ called projection maps.

Definition 7.2: Product Topology

If $\{X_\alpha\}_{\alpha \in J}$ family of top. spaces, product topology on $\prod_{\alpha \in J} X_\alpha$ is the coarsest topology s.t. all projections are continuous.

This is in general **NOT** the same as box topology.

A subbasis for product topology is collection of all sets of the form $\pi_\alpha^{-1}(U_\alpha)$ for U_α open in X_α .

Each $\pi_\alpha^{-1}(U_\alpha)$ is a set of the form $\prod_{\beta \in J} U_\beta$, where all U_β for $\alpha \neq \beta$ are X_β .

Intersection of finitely many elements in subbasis looks like $\prod_{\alpha \in J} U_\alpha$, where $U_\alpha = X_\alpha$ for all but finitely many α .

Sets of this form are a basis for product topology. No way to write a set of the form $\prod_{\alpha \in J} U_\alpha$ where $U_\alpha \subsetneq X_\alpha$ for all α as a union of these basis elements. This implies box topology is finer than product topology.

Let $\{X_\alpha\}_{\alpha \in J}$ be a family of spaces, for each $\alpha \in J$, let $f_\alpha : Y \rightarrow X_\alpha$ be a map for some spaces. Defines a map $f : Y \rightarrow \prod_{\alpha \in J} X_\alpha$ by $f(y) = (f_\alpha(y))_{\alpha \in J}$

Proposition 7.3

f is continuous with respect to product topology on $\prod_{\alpha \in J} X_\alpha \iff$ every f_α is continuous.

Proof.

$\implies$: By definition of product topology π_α is continuous, and $f_\alpha = \pi_\alpha \circ f$, so this is continuous.

$\impliedby$: Suffices to prove that $f^{-1}(U)$ is open for every U of the form $\pi_\alpha^{-1}(U_\alpha)$ for some $\alpha \in J, U_\alpha \subseteq X_\alpha$ open.

$$f^{-1}(U) = f^{-1}(\pi_\alpha^{-1}(U_\alpha)) = f_\alpha^{-1}(U_\alpha)$$

open since f_α is continuous. ■

7.1 Topologies on $\mathbb{R}^J$, where J is infinite**Proposition 7.4**

Let (X, d) be a metric spaces. The function $\bar{d} : X \times X \rightarrow \mathbb{R}_{\geq 0}$ by $\bar{d}(x, y) = \min(d(x, y), 1)$ is a metric on X , induces same topology as d .

Definition 7.5: Diameter

The Diameter of a metric space (X, d) is $\sup\{d(x, y)\}$ for all $x, y \in X$

Definition 7.6: Uniform Metric

Let J be an index set. Define metric $\bar{\rho}$ on $\mathbb{R}^J$ by

$$\bar{\rho}(x, y) = \sup_{\alpha \in J} \bar{d}(x_\alpha, y_\alpha)$$

Induces uniform topology.

When J is finite this is the same topology as product topology on $\mathbb{R}^n$

Otherwise: Uniform Topology lies strictly between the product and box topologies on $\mathbb{R}^J$.

7.1.1 Product Topology on $\mathbb{R}^\omega$

Theorem 7.7

The function $D : \mathbb{R}^\omega \times \mathbb{R}^\omega \rightarrow \mathbb{R}_{\geq 0}$ given by

$$D(x, y) = \sup_{i \in \mathbb{N}} \frac{\bar{d}(x_i, y_i)}{i}$$

is a metric on $\mathbb{R}^\omega$ which induces product topology.

Fix ball $B_D(x, r)$. Choose $N \in \mathbb{N}$ s.t. $1/N < r$. Then $B_{\bar{d}}(x_1, r) \times B_{\bar{d}}(x_2, r) \times \dots, B_{\bar{d}}(x_N, r) \times \mathbb{R} \times \mathbb{R} \times \dots$ is open in product topology. It's contained in $B_D(x, r)$, since each y in the product satisfies:

$$\bar{d}(x_i, y_i)/i \leq \bar{d}(x_i, y_i) < r \text{ if } i \leq N$$

$$\bar{d}(x_i, y_i) \leq 1/N < r \text{ if } i > N$$

8 Connectedness

Definition 8.1: Disconnected

Let X be a topological space. X is disconnected if there exists open disjoint non-empty $U, V \subseteq X$ s.t. $X = U \sqcup V$. U, V are called a separation of X .

Definition 8.2: Connected

If such a decomposition does not exist, X is connected.

Proposition 8.3

X is connected if and only if $X, \emptyset$ are the only closed and open sets

Proof.

By definition of disconnected U, V will be closed and open, so if there is not this disjoint union meaning there is no other closed and open spaces. ■

Example

A disjoint union of two or more top. spaces is disconnected $X, Y, X \sqcup Y$ is the separation. ■

Example: Connected Spaces

X with the indiscrete topology is connected.

A space containing one point is connected.

Let X be a set and $p \in X$. Let the open sets be $\emptyset$ and $A \subseteq X$ s.t. $p \in A$. X is connected.(it has no disjoint open sets at all.)

$\mathbb{R}$ and intervals and rays in $\mathbb{R}$ are connected. ■

Example: Disconnected Spaces

X with the discrete topology, $|X| \geq 2$ is disconnected.

Let $x \in X$. Then $\{x\}$ is open, and $X \setminus \{x\}$ is open. So $\{x\} \sqcup X \setminus \{x\}$ is a separation of X .

$[0, 1) \cup (1, 2] \subseteq \mathbb{R}$ is disconnected. ■

Proposition 8.4

Let $f : X \rightarrow Y$ be a continuous map and suppose X is connected. Then $f(X)$ is connected.

Proof.

Assume it is not connected s.t. $f(X) = (U \cap f(X)) \sqcup (V \cap f(X))$ for U, V open/closed in Y . Then $f^{-1}(U)$ and $f^{-1}(V)$ is open/closed. Thus, $f^{-1}(U) \sqcup f^{-1}(U)^c = X$. Thus, X is disconnected which is a contradiction. ■

Corollary 8.5

Let X be connected and let $X \rightarrow \mathbb{R}$ be a continuous map. Then if $a, b \in f(X), a < b$, the whole interval $[a, b] \subseteq f(X)$

Proof.

Let $r \in (a, b)$ and suppose $r \notin f(X)$. Then $f(X) \subseteq (-\infty, r) \cup (r, \infty)$. But in that case, $f^{-1}(-\infty, r)$ and $f^{-1}(r, \infty)$ are open and nonempty, disjoint and cover X . Which contradicts X being connected. ■

Definition 8.6: Linearly Ordered Continuum

Let L be a linearly ordered set satisfying:

- L has a least upper bound property
- If $x, y \in L, x < y \implies \exists r \in L, x < r < y$

Definition 8.7: Convex

$Y \subseteq L$ is convex if whenever $x, y \in Y$ and $r \in L, x < r < y \implies r \in Y$

Theorem 8.8

Equip L with the order topology. Then any convex subset Y of L is connected. "Intervals"

Proof.

Suppose Y is not connected s.t. $Y = A \sqcup B$, for A, B nonempty, disjoint, open in Y . WLOG let $a \in A, b \in B$ s.t. $a < b$ since Y is convex $[a, b] \subseteq Y$. Then $[a, b] \cap A$ and $[a, b] \cap B$ are open in $[a, b]$, disjoint, nonempty, so $[a, b]$ is disconnected.

Let $c = \sup[a, b] \cap A$. This exists, because $[a, b] \cap A$ is bounded by b , and sup exists by least upper bound property.

Note that $c \in [a, b]$ since $x \leq b$ for all $x \in [a, b]$ we know that $c \leq b$ thus $c \in [a, b]$

First, assume $c \in B$. Since $B \cap [a, b]$ is open in $[a, b]$, there has to exist a $d < c$ s.t. $(d, c] \subseteq B$ we can do this the fact that B is open, so we can find a tiny interval around c . But, because A is disjoint from B , $(d, c]$ is disjoint from A . Then, d is a better upper bound for $[a, b] \cap A$, so this is a contradiction.

Second, assume $c \in A$. Then because A is open we have $[c, e) \subseteq A$. By the properties of L , there is some $f \in (c, e)$. Then $f \in A \cap [a, b]$, but $f > c$. Then c is not an upper bound for $A \cap [a, b]$.

Therefore, $[a, b]$ does not have a separation, and thus is connected. ■

Lemma 8.9

If Y is a connected subspace of a space X . If C, D gives a separation of X , either $Y \subseteq C$ or $Y \subseteq D$.

Proof.

$C \cap Y$ is open in Y , $D \cap Y$ is open in Y . Then at least one of $C \cap Y$, $D \cap Y$ is empty so $Y \subseteq C \cap Y$ or $Y \subseteq D \cap Y$. This is because if they are both non empty then $Y = C \cap Y \cup D \cap Y$ then this is a contradiction of Y being connected. ■

Proposition 8.10

Suppose A connected subspace of X . Let $B \subseteq X$ be a set with $A \subseteq B \subseteq \bar{A}$. Then B is connected.

Proof.

Suppose B is not connected. Then $B = (C \cap B) \cup (D \cap B)$ for open sets C, D with $(C \cap D) \cap B = \emptyset$

Since A is connected, WLOG $A \subseteq C \cap B$ thus $\bar{A} \subseteq \overline{C \cap B}$.

Since $C \cap B \subseteq X \setminus D$ which is closed. So $\overline{C \cap B} \subseteq X \setminus D$. Thus $(\overline{C \cap B}) \cap (D \cap B) = \emptyset$.

Since $B \subseteq \bar{A} \subseteq \overline{C \cap B}$ this means that $B \cap (D \cap B) = \emptyset$, contradicts that $D \cap B, C \cap B$ is a separation of B since $D \cap B$ is empty. ■

8.1 Path Connectivity

Definition 8.11: Path

A path is a continuous map $[a, b] \rightarrow X$.

Definition 8.12: Path-Connected

X is path-connected if, for any two points $x, y \in X$, then there is a path $f : [a, b] \rightarrow X$ s.t. $f(a) = x, f(b) = y$.

Proposition 8.13

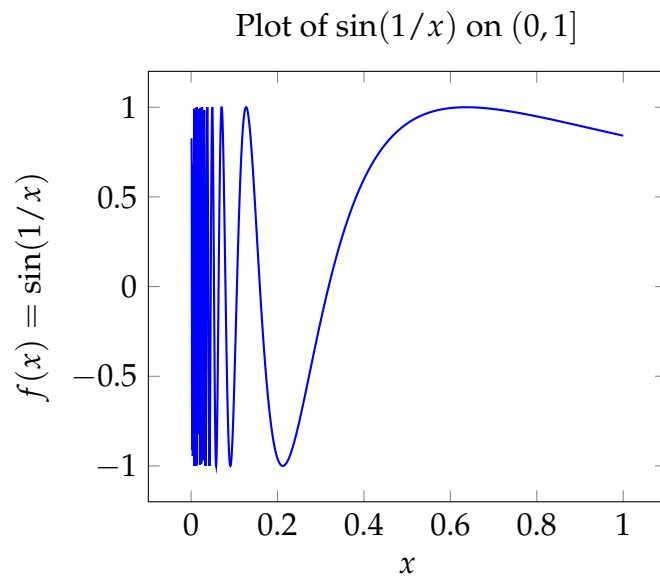
If X is path-connected, then it's connected.

Proof.

Assume X is disconnected. Let $X = A \sqcup B$ for open sets A, B

Let $x \in A$ and $y \in B$. Take any path $f : [a, b] \rightarrow X$ s.t. $f(a) = x$. The image $f([a, b])$ is a connected subspace. Thus $f([a, b]) \subseteq A$, so $f(b) \notin B$, so $f(b) \neq y$. ■

Example: Topologist's Sine Curve



$$S = \{(x, \sin(1/x)) \in \mathbb{R}^2 \mid x \in (0, 1]\}$$

S is connected since it is the graph of a continuous function. Therefore, $\bar{S}$ is also.

$$\bar{S} = S \cup (\{0\} \times [-1, 1])$$

Suppose there is a path between any point in $(\{0\} \times [-1, 1])$ to a point in S . $f : [a, b] \rightarrow \bar{S}$.

Thus, $f^{-1}(\{0\} \times [-1, 1])$ is a closed subset of $[a, b]$.

Let $m = \sup f^{-1}(\{0\} \times [-1, 1])$. This means that $f(m) \in \{0\} \times [-1, 1]$.

Can re-parametrize the interval $[m, b]$ to get a map

$f : [0, 1] \rightarrow \bar{S}$ satisfying:

$$f(0) = (0, y(0))$$

$$f(t) = (x(t), y(t)) \text{ where } x(t) > 0 \text{ whenever } t > 0$$

Find a sequence of times t_n with $t_n \rightarrow 0$ s.t.

$$y(t_n) = (-1)^n$$

Given some n , can find some $u \in \mathbb{R}$ between 0 and $x(1/n)$ s.t. $\sin(1/u) = (-1)^n$

Since $x(t)$ is continuous, $x(1/u) > u$, $x(0) < u$. By the IVT then there is a $t_n \in [0, 1/n]$ s.t. $x(t_n) = u$.

This means that $y(t_n) = \sin(1/x(t_n)) = \sin(1/u) = (-1)^n$

Conclude that $t_n \rightarrow 0$, but $f(t_n)$ does not converge. Thus f is not continuous. ■

8.2 More on Connected Spaces

Proposition 8.14

Let $\{A_j\}_{j \in J}$ be a collection of connected subspaces of a space X . If $\bigcap_{j \in J} A_j$ is nonempty, then $\bigcup_{j \in J} A_j$ is connected.

Proof.

Let $p \in \bigcap_{j \in J} A_j$, let $Y = \bigcup_{j \in J} A_j$.

Suppose for contradiction that Y is not connected. Then, there are open sets C, D s.t. $Y = (C \cup Y) \cup (D \cap Y)$ and $C \cap Y, D \cap Y$ nonempty, disjoint.

WLOG, $p \in C \cap Y$. Since each A_j is connected and is a subspace of Y we know that $A_j \subset C$ or $A_j \subset D$. But $p \in A_j \cap C$, so $A_j \subseteq C$ for all j . $Y \subset C$, so $Y \cap D = \emptyset$. $\Rightarrow \Leftarrow$ ■

Proposition 8.15

Let X, Y be connected spaces. Then $X \times Y$ is connected.

Proof.

Fix $x \in X$. Then $\{x\} \times Y$ is homeomorphic to Y , so it's connected.

For any $y \in Y$, the spaces $X \times \{y\}$ is connected.

Let $T_y = (\{x\} \times Y) \cup (X \times \{y\})$ be a subspace. It is connected since they intersect non trivially, and they are both connected.

Since $X \times Y = \bigcup_{y \in Y} T_y$. But the intersection of all the T_y is non-empty, so the unions is connected thus $X \times Y$ is connected. ■

Proposition 8.16

Let $\{X_\alpha\}_{\alpha \in J}$ be a collection of connected spaces. Then $\prod_{\alpha \in J} X_\alpha$ is connected. (w/ product topology)

Proof.

(in case where $\prod_{\alpha \in J} X_\alpha = \mathbb{R}^\omega$):

Fix the point $(0, 0, 0, \dots) \in \mathbb{R}^\omega$.

Given any N , define the subspaces $X_N \subseteq \mathbb{R}^\omega$ consisting of sequences (x_n) satisfying $x_n = 0$ for all $n > N$.

X_N is homeomorphic to $\mathbb{R}^N$. X_N is connected since $\mathbb{R}^N$ is.

$\bigcap_{N=1}^{\infty} X_N$ contains 0. Thus, the union is connected.

Claim

$$\overline{\bigcup_{N=1}^{\infty} X_N} = \mathbb{R}^\omega$$

Proof.

Let U be any open set in $\mathbb{R}^\omega$, where $U = U_1 \times U_2 \times \dots \times U_n \times \mathbb{R} \times \mathbb{R} \times \dots$

This open set U intersect $\bigcup_{N=1}^{\infty} X_N$:

there is a point in U of the form $(x_1, x_2, \dots, x_m, 0, 0, \dots)$

This point lies in X_M . So every nonempty open set intersects the union, so its closure is $\mathbb{R}^\omega$

Thus since the closure of a connected space is connected $\mathbb{R}^\omega$ is connected.

Definition 8.17: Connected Component

Let X be a space. A connected component of X is a maximal connected subspace of X . If A is connected and $A \supseteq X$ and A is connected, then $A = X$

Example

$X = [0, 1/2) \cup (1/2, 1]$, each of $[0, 1/2)$ and $(1/2, 1]$ is a connected component.

Define an equivalence relation $\sim$ on X by:

$x \sim y$ if there is a connected subspace of X containing both x and y .

Proposition 8.18

Connected components are equivalence classes under $\sim$.

Proof.

Every connected subspace of X intersects exactly one equivalence class. If C_1, C_2 are equivalence class, and $x \in A \cap C_1, y \in A \cap C_2$ for connected set A , then $x \sim y$ so $C_1 = C_2$

So, equivalence classes are connected:

Fix $x_0 \in C$ for equivalence class C . For each $y \in C$ there is a connected A_y containing x_0, y . Know $A_y \subseteq C$, so $\bigcup_{y \in C} A_y = C$.

$\bigcap_{y \in C} A_y$ contains x_0 so C is connected.

If A connected, contains equivalence class C , by def every $z \in A$ is equivalent to every $y \in C$, so $A \subseteq C$ and $A = C$.

X is connected $\iff X$ has exactly one or zero connected components. ■

Connected components are closed, but not necessarily open.

Example

Connected Components of $\mathbb{Q}$ are all singletons. ■

Definition 8.19: Totally Disconnected

A space X is totally disconnected if every point is a connected component.

Definition 8.20: Path Component

Let X be a space. A path component of X is a maximal connected path of X . If A is path and $A \supseteq X$ and A is connected, then $A = X$

9 Compactness

Definition 9.1: Compact

A space X is compact if every open covering $\mathcal{C}$ of X has a finite sub cover, i.e. $\exists$ sets $U_1, \dots, U_n \in \mathcal{C}$ s.t. $X = U_1 \cup \dots \cup U_n$

Example: $\mathbb{R}$ is not compact. Also, any unbounded subspace of a metric space is not compact.

(X, d) metric space, $Y \subseteq X$ unbounded, then $\{B(x, n) \mid n \in \mathbb{N}\}$ is an open conering of Y , but you cannot pick a finite among since you can always find bigger things in Y that are not in your finite set. ■

Example: $X = \{0\} \cup \{1/n \mid n \in \mathbb{N}\}$ is compact

Let $\mathcal{C}$ be an open covering. Then $0 \in U$ for $U \in \mathcal{C}$.

U is open, so it contains an interval $(-\epsilon, \epsilon)$. Since $1/n \rightarrow 0$ we know that for all but finitely many n , $1/n \in (-\epsilon, \epsilon)$. Then take the reaming not inside that ball, and you can just pick an open set for each remaining point. ■

Proposition 9.2

Every closed subset of a compact space is compact.

Proof.

Let X be a compact, let $Y \subseteq X$ be closed.

Fix an open covering $\mathcal{C}$ of Y

Consider family of sets $\mathcal{C} \cup \{X \setminus Y\}$.

$X \setminus Y$ is open, so this is an open covering of X .

X is compact $\implies$ has a finite subcover, call it $\mathcal{C}' \cup \{X \setminus Y\}$, $\mathcal{C}' \subseteq \mathcal{C}$ finite. $\mathcal{C}'$ finite open covering of Y , subcover of $\mathcal{C}$ ■

Theorem 9.3

Let X be Hausdorff space. If $Y \subseteq X$ is compact, then it's closed in X .

Proof.

Let $x_0 \in X \setminus Y$.

For each $y \in Y$, $\exists U_y$ open of x_0 , V_y of y s.t. $U_y \cap V_y = \emptyset$

Look at the collection $\{V_y\}_{y \in Y}$. This is an open covering of Y , has finite subcover

$V_{y_1}, V_{y_2}, \dots, V_{y_n}$.

$V_{y_i} \cap U_{y_i} = \emptyset$, so if $V = V_{y_1} \cup V_{y_2} \cup \dots \cup V_{y_n}$ and $U = U_{y_1} \cap U_{y_2} \cap \dots \cap U_{y_n}$ thus $U \cap V = \emptyset$.

Since U is open, and V contains Y , so $U \cap Y = \emptyset$, U is open subset contained in $X \setminus Y$, and U contains x_0 thus $X \setminus Y$ is open. ■

Theorem 9.4

If X compact, and $f : X \rightarrow Y$ is continuous, then $f(X)$ is compact.

Proof.

Let $\mathcal{C}$ be an open covering of $f(X)$. Consider $\{f^{-1}(U) \mid U \in \mathcal{C}\}$ is an open covering of X , thus there is a finite amount $U_1, U_2, \dots, U_n$ in $\mathcal{C}$ s.t. $f^{-1}(U_1), \dots, f^{-1}(U_n)$ covers X . Then $U_1, \dots, U_n$ covers $f(X)$. ■

Theorem 9.5

Let $f : X \rightarrow Y$ be a continuous bijection.

If X is compact, and Y is Hausdorff, then f is a homeomorphism.

Proof.

Let $g : Y \rightarrow X$ be inverse map. Let $A \subset X$ be closed. Thus, A is compact. Thus, $f(A)$ is compact, since $g^{-1}(A) = f(A)$. We know that since Y is Hausdorff that $g^{-1}(A) = f(A)$ is closed. Therefore, g is continuous. ■

Corollary 9.6

If X is compact, and $f : X \rightarrow \mathbb{R}$, then f is bounded.

Proof.

Then we know that $f(X)$ is compact, thus there is a finite amount of intervals that can cover, so there is a finite radius thus f is bounded. ■

Definition 9.7: Proper

A metric space (X, d) is proper if every closed ball in X is compact.

9.1 Heine-Borel Theorem

Theorem 9.8: Heine-Borel Theorem

Let $Y \subseteq \mathbb{R}^n$. Then Y is compact $\iff Y$ is closed and bounded

Proposition 9.9

Let A be a compact subset of a metric space. Then A is closed and bounded

Proof.

Closed because space is Hausdorff. Bounded because ball $B(x_0, n)$ give an open covering which then has a finite subcover, so it is bounded by the biggest ball in the subcover. ■

Theorem 9.10

Let X be any ordered set with least upper bound property. Then any closed interval $[a, b] \subseteq X$ is compact.

Proof.

In case $X = \mathbb{R}$

Let $\mathcal{C}$ be an open covering of closed interval $[a, b]$

Claim: 1

For any $x \in [a, b)$, $\exists y > x$ in $[a, b]$ s.t. $[x, y]$ is contained in an element of $\mathcal{C}$

Proof.

Let $C \in \mathcal{C}$ contain x .

C is open, so it contains an interval $[x, c)$, for $c > x$, $c \in [a, b]$.

Let $y \in (x, c)$. Then C contains $[x, y]$ ■

Let A be set of all points $y \in [a, b]$ s.t interval $[a, y]$ can be covered by finitely many elements in $\mathcal{C}$

A is nonempty, and contains a point $> a$ by the claim above.

Let $s = \sup A \subseteq [a, b]$. This is an element in $(a, b]$.

Claim: 2

$s \in A$

Proof.

Let $C \in \mathcal{C}$ contains s . C contains an interval $(d, s]$, for $d < s$.

$(d, s) \cap A$ is nonempty. (if it is empty then d would be an upper bound for A).

Let $z \in (d, s) \cap A$, then $[z, c]$ is in C . Since $z \in A$, $[a, z]$ is covered by finitely many elements of $\mathcal{C}$

Thus, $[a, s]$ is in the unions of fin. many elts. in $\mathcal{C}$, so $s \in A$. ■

If $s < b$, by Claim 1, $\exists y \in (s, b]$ s.t. $[s, y]$ is in a single element in $\mathcal{C}$.

$[a, s]$ is covered by fin. many elts. of $\mathcal{C}$, $[s, y]$ is in one element of $\mathcal{C}$, so $[a, y]$ is in finitely many elts. of $\mathcal{C}$ thus $y \in A$ contradicts $s = \sup A$.

Thus $A = [a, b]$, thus $[a, b]$ is compact. ■

Lemma 9.11

Suppose X, Y spaces, Y compact. Let $N \subseteq X \times Y$ be an open set containing $\{x_0\} \times Y$. Then there exists an open set W in X s.t. $\{x_0\} \times Y \subseteq W \times Y \subseteq N$.

Proof.

Know N is a union of sets of the form $U \times V$, for U, V open.

$\{x_0\} \times Y$ is covered by sets of this form, contained in N , since $\{x_0\} \times Y \cong Y$ it will be contained in a finite union

$$(U_1 \times V_1) \cup \dots \cup (U_n \times V_n)$$

where $U_i \times V_i$ is in N .

WLOG every $U_i \times V_i$ intersects $\{x_0\} \times Y$. This means that each U_i contains x_0 , so the intersection of U_i 's called W is an open set containing x_0 .

Then $\{x_0\} \times Y \subseteq W \times Y$.

Let $(x, y) \in W \times Y$. Then $(x_0, y) \in \{x_0\} \times Y$. So $(x_0, y) \in U_i \times V_i$ for some i , so $(x, y) \in U_i \times V_i$ since x is in every U_i .

Thus $\{x_0\} \times Y \subseteq W \times Y \subseteq N$ ■

Theorem 9.12

Let X, Y be compact spaces. Then $X \times Y$ is compact.

Proof.

Let X, Y be compact, let $\mathcal{C}$ be an open covering of $X \times Y$.

Fix $x \in X$. Since $\{x\} \times Y$ is compact, it is covered by finitely many sets $U_1, \dots, U_n$ in $\mathcal{C}$.

Let N be the unions of all the U_i 's then by above there is a tube $W_x \times Y \subseteq N$.

Thus, $W_x \times Y$ is in finitely many sets in $\mathcal{C}$.

Since X is compact, and the sets $\{W_x\}_{x \in X}$ cover X , can find finitely many tubes $W_{x_1} \times Y, \dots, W_{x_n} \times Y$ s.t. $W_{x_1}, \dots, W_{x_n}$ cover X .

This collection of tubes is a cover of $X \times Y$, and each tube is covered by finitely many sets in my cover, thus $X \times Y$ is covered by finitely many sets in $\mathcal{C}$. ■

Proof.

Pf of Heine-Borel

($\implies$): done by proposition.

($\impliedby$): If A bounded, it is contained in a box. $[-M, M]$. This is compact by theorem above thus $[-M, M]^n$ is compact. A is a closed subset of a compact space thus A is compact. ■

9.2 More on Compactness

Theorem 9.13: Extreme Value Theorem

Let X be a compact space. Let $f : X \rightarrow \mathbb{R}$ be a continuous function. Then f achieves its minimum and maximum values. i.e. $\exists a, b \in X$ s.t. $f(a) \leq f(x) \leq f(b)$ for all $x \in X$

Proof.

$f(X)$ is a compact subset of $\mathbb{R}$. It is closed and bounded. Let $s = \sup f(X)$

Consider the family of intervals $\{(-\infty, t) \mid t < s\}$

If $s \notin f(X)$

This is an open covering of $f(X)$ which has no finite subcover, since

Definition 9.14: Uniformly Continuous

Let $(X, d_X), (Y, d_Y)$ be metric spaces.

A function $f : X \rightarrow Y$ is uniformly continuous if $\forall \epsilon > 0, \exists \delta > 0$ s.t.

$$d_X(x_1, x_2) < \delta \implies d_Y(f(x_1), f(x_2)) < \epsilon$$

Theorem 9.15: Uniform Continuity Theorem

Let X, Y be metric spaces, and suppose X is compact.

Then any continuous function $f : X \rightarrow Y$ is uniformly continuous.

Theorem 9.16

Let $\mathcal{C}$ be an open covering of a compact metric space X . Then there is some $\delta > 0$ s.t. every subset of X with diameter $< \delta$ is contained in some set $C \in \mathcal{C}$

δ is called a Lebesgue number for the cover $\mathcal{C}$

Proof.

For each $x \in X$, let C_x be a set in $\mathcal{C}$ containing x . There is a ball $B(x, r_x) \subset C_x$ for some $r_x > 0$.

Consider the open covering of X given by the set of balls

$$\{B(x, \frac{r_x}{2}) \mid x \in X\}$$

X is compact, so take a finite subcover

$$B(x_1, \frac{r_1}{2}), \dots, B(x_n, \frac{r_n}{2})$$

Let $\delta = \min(\frac{r_1}{2}, \dots, \frac{r_n}{2}) > 0$

Let $A \subset X$ have diameter $< \delta$. Then for $z \in A$ is contained in a ball $B(x_i, \frac{r_i}{2})$.

For any $y \in A$, $d(y, z) < \delta$, so $d(x_i, y) < d(x_i, z) + d(z, y) < \frac{r_i}{2} + \delta \leq r_i$.

Thus $A \subseteq B(x_i, r_i) \subseteq C_{x_i}$ ■

Proof.

[Uniform Continuity Theorem]:

Let $\epsilon > 0$. Consider open covering of Y given by

$$\{B(y, \frac{\epsilon}{2}) \mid y \in Y\}$$

The sets

$$\{f^{-1}(B(y, \frac{\epsilon}{2})) \mid y \in Y\}$$

give an open covering of X .

Let δ be a Lebesgue number for this covering.

If $x_1, x_2 \in X$ s.t. $d(x_1, x_2) < \delta$, $\text{diam}\{x_1, x_2\} < \delta$, so x_1, x_2 both lie in some

$$f^{-1}(B(y, \frac{\epsilon}{2})) \implies f(x_1), f(x_2) \in B(y, \frac{\epsilon}{2}) \implies d(f(x_1), f(x_2)) < \epsilon. \quad \text{■}$$

9.2.1 The Cantor Set

Let $C_0 = [0, 1]$

Let $C_1 = C_0$ minus middle third of interval

Let $C_2 = C_1$ minus middle third of each component interval of C_1

In general,

C_i is a union of closed intervals

C_{i+1} is the set obtained by deleting "middle third" interval of each connected component of C_i

The Cantor set is $C = \bigcap_{i=0}^{\infty} C_i$

No endpoint of an interval in any C_i is deleted. In particular, $0, 1$ all in C , $3^{-n} \in C$ always.

Each C_i is closed in $\mathbb{R}$ finite union of closed intervals thus C is closed, and bounded by $0, 1$, so it is compact.

It contains many embedded copies of itself.

Definition 9.17: Isolated

A point x in a space X is called isolated if $\{x\}$ is open in X .

Cantor set contains no isolated points.

Theorem 9.18

Let X be any compact Hausdorff space.

If X has no isolated points, then it is uncountable.

C shows up all the time.

Let $\{X_j\}_{j \in \mathbb{N}}$ be a family of spaces s.t. each X_j is finite, discrete, and non-empty.

$$\prod_{j \in \mathbb{N}} X_j$$

is always homeomorphic to Cantor set.

$\{0, 1\}^\omega$ is homeomorphic to Cantor set.

Can specify a point in Cantor set by specifying which components of each C_i it belongs to, same as a sequence of left right choices.

Theorem 9.19

Let X be any space satisfying the following:

- X is compact
- X is metrizable
- X has no isolated points
- X is totally disconnected

X is homeomorphic to the Cantor set.

Theorem 9.20

Let X be any compact metrizable space. Then X is homeomorphic to a quotient $C / \sim$ where C is the Cantor set.

9.3 Sequential Compactness

Definition 9.21: Subsequence

Let x_n be a sequence in some X .

If $n_1 < n_2 < n_3 < \dots$ is a sequence of strictly increasing indices, then the sequence $x_{n_1}, x_{n_2}, \dots$ is a subsequence of x_n .

Definition 9.22: Sequentially Compact

A topological space X is sequentially compact if every sequence of X has a convergent subsequence.

Theorem 9.23

Suppose X is a metrizable space. Then X is compact if and only if X is sequentially compact.

Proof.

($\implies$)

Suppose X is not sequentially compact, then let x_n be a sequence with no convergent subsequence.

Let A be the set of elements appearing in the sequence x_n .

Claim

A is closed

Proof.

Let $x \in \bar{A}$. Then, because X is metrizable, x is a limit of a sequence of elements $y_n \in A$.

If $x \notin A$, then a subsequence of y_n is also a subsequence of x_n .

Know $y_1 = x_{n_1}$. Assume,

$$y_{m_1} = x_{n_1}, \dots, y_{m_j} = x_{n_j}$$

for indices

$$m_1 < m_2 < \dots < m_j$$

$$n_1 < n_2 < \dots < n_j$$

The set $\{y_m \mid m > m_j\}$ is infinite since if it were finite we would get a constant subsequence which implies that $x \in A$.

This intersects $A \setminus \{x_1, \dots, x_{n_j}\}$.

So take $y_{m_{j+1}} = x_{n_{j+1}}$ to be any elt in this set.

Build a subseq. of y_n converging with subsequence of x_n . This is a contradiction thus $x \in A \implies \bar{A} = A$ thus A is closed. ■

For each m we know x_m is not the limit of a subsequence of x_n . So, there exists a neighborhood of x_m which contains only finitely many elements from A .

Also know X is Hausdorff, so there is a neighborhood U_n of x_n s.t. U_n contains no other elements of A .

Covering:

$$(X \setminus A) \sqcup \bigcup_{n=1}^{\infty} U_n$$

has no finite subcover. Thus, X is not compact.

($\Leftarrow$) Assume X is sequentially compact and metrizable.

Claim: 1

Any open covering of X has a Lebesgue number.

Proof.

Let $\mathcal{C}$ be an open cover of X . Assume for contradiction no Lebesgue number exists.

So, for all $n \in \mathbb{N}$ there is a set D_n s.t. $\text{diam}(D_n) < 1/n$, and $D_n \not\subseteq C$ for any $C \in \mathcal{C}$.

Fix $x_n \in D_n$. Let x be the limit of a subsequence of x_n . $x \in C \subseteq \mathcal{C}$, so $x \in B(x, \epsilon) \subseteq C$ for some $\epsilon > 0$.

For some large n , we have $d(x, x_n) < \epsilon/2$ and $1/n < \epsilon/2$.

$x_n \in D_n$, and $\text{diam}(D_n) < 1/n < \epsilon/2$ by triangle inequality, $D_n \subseteq B(x, \epsilon) \subseteq C$. This is a contradiction. ■

Claim: 2

Given any $\epsilon > 0$, there exists a finite collection of points $x_1, \dots, x_n$ s.t.

$$\bigcup_{i=1}^n B(x_i, \epsilon) = X$$

Proof.

Assume for contradiction that the claim is not possible. Fix $x_1 \in X$. $B(x_1, \epsilon) \neq X$, so let $x_2 \in X \setminus B(x_1, \epsilon)$.

Now define sequence s.t. $x_i \notin B(x_1, \epsilon) \cup \dots \cup B(x_{i-1}, \epsilon)$.

We know that the union of these j balls is not equal to X , so let x_{j+1} be in the complement of the union of the j balls.

This sequence x_n has no convergent subsequence since $d(x_i, x_j) > \epsilon$ for all $i \neq j$.

This means that no subsequence is Cauchy. Thus, no subsequence converges. ■

Let $\mathcal{C}$ be an open covering of X . Let $\delta > 0$ be a Lebesgue number for the covering. Cover X by finitely many balls with radius $\epsilon = \delta/3$.

Thus, $\text{diam}(B(x, \epsilon)) \leq 2\epsilon = 2\delta/3 < \delta$, so each ball in this finite covering is contained in some $C \in \mathcal{C}$.

Pick some C containing each ball in the cover by ϵ -balls, so this is a finite subcover of X by sets in $\mathcal{C}$. ■

Theorem 9.24: Well-Ordering theorem

Every set can be equipped with a well-ordering

Corollary 9.25

There exists an uncountable well-ordered set.

Theorem 9.26

There exists a well-ordered uncountable set ω_1 s.t. every interval $(-\infty, a) \subset \omega_1$ is countable $[0, a)$

Proof.

Let S be any uncountable well-ordered set

Claim

There exists another uncountable well-ordered set T which contains an uncountable interval

Proof.

Let $T = S \times \{1, 2\}$ with dictionary order. ■

The interval $[0, z)$ is uncountable if $z = (b, 2)$ for $b \in S$.

Let Ω smallest element $z \in T$ s.t. $[0, z)$ is uncountable.

Define $\omega_1 = [0, \Omega)$, $\overline{\omega_1} = [0, \Omega]$

Every interval in $[0, a)$ in ω_1 is countable. ■

Proposition 9.27

Every countable subset of ω_1 is bounded.

Proof.

Let $A \subset \omega_1$ be countable. Let $B = \bigcup_{a \in A} [0, a]$.

B is countable, so $B \subsetneq \omega_1$. $\exists z \in \omega_1 \setminus B$.

$B \subseteq [0, z)$ if $\exists y \in B, y \geq z$, then $[0, y] \subset B$, but $y \in [0, y]$ and $z \notin B$.

Thus, B is upper bounded since $\overline{\omega_1}$ is well-ordered, so it satisfies least upper bound property. Thus, it is closed and bounded, so it is compact.

ω_1 is not compact. ■

10 Local Properties of Topological Spaces

10.1 Locally Connectivity

Definition 10.1: Locally Connected

Let X be a space. Then X is locally connected at a point $x \in X$ if, for every neighborhood U of x , there is a neighborhood $V \subseteq U$ of x s.t. V is connected.
 X is locally connected if it's locally connected at every point.

Definition 10.2: Locally Path Connected

Let X be a space. Then X is locally path connected at a point $x \in X$ if, for every neighborhood U of x , there is a neighborhood $V \subseteq U$ of x s.t. V is path connected.
 X is locally path connected if it's locally connected at every point.

Local connectedness neither implies nor is implied by connectedness.

Example

- $\mathbb{R}$ is connected, also locally connected.
- $[0, 1) \cup (1, 2]$ is locally connected.
- Topologist's sign curve is connected but not locally connected.
- $\mathbb{Z}$ is not connected, but locally connected
-

Theorem 10.3

If X is locally connected, then connected components are closed and open.

Proof.

Let C be a connected component of X .

If $x \in C$, then $\exists$ connected neighborhood V of x s.t. $V \subseteq C$.

Thus, $C = \bigcup_{x \in C} V$ so C is open.

Same proof works if you replace locally connected with locally path connected.

Theorem 10.4

If X is a topological space and X is locally path-connected, then connected components of X are the same as the path components.

Proof.

Let C be a component. If P is any path component intersecting C , then $P \subseteq C$ since path connected space is connected.

This means that:

$$C = \bigcup_{P \text{ path component intersecting } C} P$$

If there is more than one path component intersecting C , then

$$C = P \cup \bigcup_{Q \text{ path component intersecting } C} Q$$

This is a contradiction of C being connected since path components are open in locally path connected spaces. ■

10.2 Local Compactness

Definition 10.5: Locally Compact

X is locally compact at x if there is a neighborhood U of x and a compact set C containing U .

X is locally compact if it's locally compact of every $x \in X$.

Example

- Any compact space is locally compact.
 - $\mathbb{R}^n$
 - $\mathbb{Q}$ is not locally compact.
 - $\mathbb{R}^\omega$ is not locally compact.
-

Definition 10.6: Compactification

A compactification of a space X is a space Y , containing X as a subspace, s.t.:

- Y is compact and Hausdorff
- $\overline{X} = Y$

Definition 10.7: One-point Compactification

Let X be a space and let Y be its compactification then if $Y \setminus X$ is a singleton then Y is called the one-point compactification of X .

Example

- S^1 is one-point compactification of $\mathbb{R}$.
- $[0, 1]$ is a compactification of $\mathbb{R}$.
- $\{0\} \cup \{1/n \mid n \in \mathbb{N}\}$ is one-point compactification of $\mathbb{N}$.

Theorem 10.8

X has a one-point compactification $\iff X$ is locally compact and Hausdorff.

Theorem 10.9

If Y, Y' are two one-point compactifications of X , then there is a homeomorphism $Y \rightarrow Y'$ restricting to identity map.

Corollary 10.10

If X is Hausdorff, then X is locally compact $\iff$ for every $x \in X$ and every neighborhood U of x , there exists a neighborhood V of x s.t. $\overline{V}$ is compact, and $\overline{V} \subseteq U$.

Proof.

$\Leftarrow$ Just take V has your neighborhood which is contained in $\overline{V}$

$\Rightarrow$: Suppose X is locally compact and Hausdorff. Let Y be the one point compactification. Let U be open in X , $x \in U$. $Y \setminus U$ is closed, so $Y \setminus U$ is compact.

$\{x\}$ is compact, by hw problem there are open sets V, W s.t. $x \in V$, $Y \setminus U \subseteq W$, and $V \cap W = \emptyset$.

$Y \setminus W$ is closed thus compact, $V \subset Y \setminus W \subset U$ so $\overline{V} \subseteq Y \setminus W$, so $\overline{V}$ is closed subset of compact space thus $\overline{V}$ is compact. ■

Proposition 10.11

Let Y be one point compactification of X , let Y' be any compactification of X . There is a unique quotient map $Y' \rightarrow Y$ which restricts to identity on X , taking $Y' \setminus X \rightarrow Y \setminus X$.

Proof.

Let $h : Y' \rightarrow Y$ by: h is identity on X , $h(y) = \infty$ for any $y \in Y' \setminus X$. $Y = X \cup \{\infty\}$.

Let U be any open set in Y .

If $U \subseteq X$, then $h^{-1}(U)$ open $\iff U$ open, since h is identity in X

If $\infty \in U$ then U open $\iff Y \setminus U$ closed $\iff Y \setminus U$ in $X \iff Y' \setminus h^{-1}(U)$ compact $\iff Y' \setminus h^{-1}(U)$ closed $\iff h^{-1}(U)$ open. ■

Corollary 10.12

One point compactifications are unique.

Example

- S^2 is one point compactification of $\mathbb{R}^2$
- Closed unit disk is a compactification of $\mathbb{R}^2$, so we know there is a quotient map from here to S^2 .

For Theorem 10.8 *Proof.*

Let X be locally compact and Hausdorff.

Let $Y = X \cup \{\infty\}$

Define a topology $\mathcal{T}$ on Y by taking two types of open sets:

- all open subsets of X are in $\mathcal{T}$.
- all open sets of the form $Y \setminus C$, where C is a compact subset of X .

Show $\mathcal{T}$ is a topology.

Show Y is compact.

Show Y is Hausdorff. ■

11 Categorizing Spaces

Definition 11.1: First Countable

A space X is first-countable if for every, $x \in X$, there exists a countable neighborhood basis of x : There exists a collection β of neighborhoods of x s.t. β is countable, and every neighborhood of x contains some $B \in \beta$

Proposition 11.2

Let X be first countable. Then $f : X \rightarrow Y$ is continuous if and only if for all sequences $x_n \rightarrow x$, we have $f(x_n) \rightarrow f(x)$.

Proposition 11.3

Let X be first countable, let $A \subseteq X$. Then $x \in \overline{A}$ if and only if there exists x_n in A s.t. $x_n \rightarrow x$.

Definition 11.4: Second Countable

A space X is second countable if it has a countable basis.

Proposition 11.5

Let X be a second countable space. Then any discrete subspace of X is countable.

Proof.

Consider $A \subseteq X$ discrete subspace.

For each $x \in A$, can find a basis element B_x s.t. $B_x \cap A = \{x\}$.

So, if β is a countable basis, the map $A \rightarrow \beta$ give by $x \mapsto B_x$ is injective map from A to a countable set thus A is countable. ■

Proposition 11.6

Any subspace of a first countable space is first countable.

Proposition 11.7

Any countable product of a first countable space is first countable.

Also works for second.

Proposition 11.8

Let X be second countable. Then X has a countable dense subset.

Proof.

Let $\{B_n\}$ be a countable basis for X . For each B_n , let $x_n \in B_n$.

Define: $A = \{x_n\}$.

Let $x \in X$. Then any neighborhood U of x contains some B_n , so it contains some $x_n \in A$.

Thus, $x \in \overline{A}$. ■

Definition 11.9: Separable

X is separable if it contains a countable dense subset.

Definition 11.10: n -Manifold

Let X be a topological space, $n \in \mathbb{N}$. Then X is an n -manifold if:

- it is Hausdorff
- it is second countable
- it is locally homeomorphic to $\mathbb{R}^n$

X is locally homeomorphic to $\mathbb{R}^n$ if for all $x \in X$, there is a neighborhood U of x which is homeomorphic to an open subset of $\mathbb{R}^n$

A 1-manifold is a curve, and a 2-manifold is a surface.

Theorem 11.11

Every connected 1-manifold is homeomorphic to either S^1 or $\mathbb{R}$.

Theorem 11.12

Any n -manifold is a metrizable space.

Theorem 11.13

Any n -manifold can be embedded into a copy of $\mathbb{R}^N$, for some $N \geq n$.

12 Separation Axioms

Definition 12.1

Let X be a topological space

- X is T_1 if singletons are closed.
- X is Hausdorff (T_2) if, for every pair of distinct points $x, y \in X$, $\exists$ disjoint neighborhoods of x, y
- X is regular (T_3) if it is T_1 and for every $x \in X$ and every closed $B \subset X$ with $x \notin B$, $\exists$ disjoint open sets containing x and B .
- X is normal if it is T_1 and for every pair of disjoint closed subsets $A, B \subset X$, $\exists$ disjoint open sets containing A, B .

Theorem 12.2

A subspace of a T_1, T_2, T_3 space is T_1, T_2, T_3

Theorem 12.3

Let $\{X_\alpha\}_{\alpha \in J}$ be a family of T_1, T_2, T_3 spaces. Then the product is T_1, T_2, T_3 .

Lemma 12.4

Let X be a top. space. Then X is regular $\iff$ for every $x \in X$ and any neighborhood U of x , there exists a neighborhood V of x s.t. $\bar{V} \subset U$.

Proof.

($\implies$) Suppose X is regular. Fix $x \in X$, neighborhood U of x .

Since $X \setminus U$ is closed, so can find disjoint open sets containing $x, X \setminus U$.

$X \setminus U \subset U$ and $x \in V$.

$X \setminus W$ is closed set containing V , so $\bar{V} \subset X \setminus U$, and since $X \setminus U \subset W$, we have $X \setminus W \subset U$, so $\bar{V} \subset U$.

($\impliedby$)

Let $x \in X, B \subset X$ s.t. $x \notin B$, assume B is closed.

Consider $U = X \setminus B$ is a neighborhood of x . Let V be a neighborhood of x s.t. $\bar{V} \subset U$.

Thus, $V, X \setminus \bar{V}$ are open sets separating x and B . ■

Theorem 12.5

If X is a metrizable space, X is normal.

Proof.

Let A, B be disjoint closed sets subsets of X .

For each $a \in A$, there exists an $\epsilon_a > 0$ s.t. $B(a, \epsilon_a) \cap B = \emptyset$.

For each $b \in B$, there exists an $\epsilon_b > 0$ s.t. $B(b, \epsilon_b) \cap A = \emptyset$.

Consider $B(a, \epsilon_a/2), B(b, \epsilon_b/2)$

If $z \in B(a, \epsilon_a/2) \cap B(b, \epsilon_b/2)$, then $d(a, b) \leq (\epsilon_a + \epsilon_b)/2 \leq \max(\epsilon_a, \epsilon_b)$. Which is a contradiction.

Thus,

$$\bigcup_{a \in A} B(a, \epsilon_a/2) \quad \bigcup_{b \in B} B(b, \epsilon_b/2)$$

are disjoint open sets containing A, B . ■

Theorem 12.6

Every compact Hausdorff spaces is normal.

Proof.

Use homework since closed means compact in compact Hausdorff. ■

Corollary 12.7

Manifolds are regular.

Proof.

Let M be a n -manifold. By definition, M is Hausdorff.

M is locally homeomorphic to $\mathbb{R}^n$, and $\mathbb{R}^n$ is locally compact, so M is locally compact.

Thus, M has a 1-point compactification. $\hat{M}$.

$\hat{M}$ is a compact Hausdorff space. Thus, $\hat{M}$ is normal. Thus, regular, and subspaces of regular spaces are regular. ■

Theorem 12.8

Every second countable regular space is normal.

Proof.

Let β be a countable basis for X .

Let A, B be disjoint closed subsets of X .

Fix $a \in A$. Then there is a neighborhood U of a that is disjoint from B .

By regularity there is a neighborhood V of a s.t. $\overline{V} \subset U$.

Thus, there is a $W \in \beta$ s.t. $a \in W \subset U$.

Then do this for any point $a \in A$, and thus we get a countable open covering of A by sets whose closures are disjoint from B .

Do the same for B .

Call them $\{U_n\}$ and $\{V_n\}$.

Then for all n define:

$$U'_n = U_n \setminus \bigcup_{i=1}^n \overline{V_n}$$

$$V'_n = V_n \setminus \bigcup_{i=1}^n \overline{U_n}$$

Thus, this union is disjoint open sets that contain A, B . ■

Theorem 12.9: Urysohn lemma

Let X be a normal space, let A, B be closed subsets. Then there exists a continuous function. $f : X \rightarrow [0, 1]$ s.t. $f(x) = 0$ for all $x \in A$, $f(x) = 1$ for all $x \in B$.

Theorem 12.10: Urysohn Metrization Theorem

Every regular space with a countable basis is metrizable

Proof.

By Theorem 12.8 we know that this space is normal.

Let X be normal and second countable.

Claim

There exists countable collection of continuous functions $f_n : X \rightarrow [0, 1]$ s.t. for every point $x \in X$, and every neighborhood U of x . $\exists n$ s.t. $f_n(x) = 1, f_n(X \setminus U) = \{0\}$

Proof.

Let β be a countable basis.

For each pair of indices n, m s.t. $\overline{B_n} \subset B_m$, define function $f_{n,m} : X \rightarrow [0, 1]$ s.t. $f_{n,m}(\overline{B_n}) = \{1\}$, $f_{n,m}(X \setminus B_n) = \{0\}$

The $f_{n,m}$ are indexed by a subset of $\mathbb{N} \times \mathbb{N}$, so this is a countable collection.

Fix $x \in X$, neighborhood U of x , $\exists B_n$ s.t. $x \in B_n \subset U$

By regularity, $\exists$ neighborhood V of x s.t. $\overline{V} \subset B_n$.

Thus, $\exists B_m$ s.t. $x \in B_m \subset V$. Thus, $x \in B_m$ and $\overline{B_m} \subset B_n$.

Thus, $f_{n,m}(x) = 1$, $f_{n,m}(X \setminus U) \subset f_{n,m}(X \setminus B_m) = \{0\}$.

Reindex to the collection $\{f_n\}$

Define $F : X \rightarrow \mathbb{R}^\omega$ s.t. $F(x) = (f_1(x), f_2(x), f_3(x), \dots)$

F is continuous since each component is continuous.

Injective:

Let $x, y \in X$ s.t. $x \neq y$. X is Hausdorff, so there exists a neighborhood U of x s.t. $y \notin U$.

Thus, there exists n s.t. $f_n(x) > 0$ and $f_n(z) = 0$ for all $z \notin U$, so $f_n(y) = 0 \implies f_n(x) \neq f_n(y) \implies F(x) \neq F(y)$.

Let $Z = F(X)$.

Let U be open in X . Consider $F(U)$.

Let $z \in F(U)$ thus there exists $x \in U$ s.t. $F(x) = z$. Thus, there is a n s.t. $f_n(x) > 0$ and $f_n(X \setminus U) = \{0\}$.

Consider the index- n projection map $\pi_n : \mathbb{R}^\omega \rightarrow \mathbb{R}$.

Consider the set $V = \pi_n^{-1}((0, \infty)) \subset \mathbb{R}^\omega$.

$f_n(x) > 0 \implies f(x) = z \in V$.

Let $y \in V \cap Z$ then $y = F(c)$ for some $c \in X$, so $f_n(c) > 0$. But then $c \in U$, since $f_n(X \setminus U) = \{0\}$.

Thus, $V \cap Z \subset F(U)$, so $F(U)$ is open. Thus, F is an embedding into $\mathbb{R}^\omega$. Thus, X is homeomorphic to a subspace of a metric space, so X is a metric space.

Corollary 12.11

Let Y be a space which Hausdorff, locally compact, and second countable. Then Y is metrizable.

Proof.

Y is Hausdorff and locally compact, so it has a one point compactification so $\hat{Y}$ is compact and Hausdorff so normal. Thus, Y is regular, and by the theorem it is metrizable.

12.1 Partition of unity

Definition 12.12: Support

Let X be any spaces, let $\phi : X \rightarrow \mathbb{R}$ be continuous. The support of ϕ is the closure of the set $\phi^{-1}(\mathbb{R} \setminus \{0\})$.
 $\text{supp}(\phi)$

Definition 12.13: Partition of unity subordinate to U

Let $U = \{U_1, \dots, U_k\}$ be a finite open covering of a space X . A partition of unity subordinate to U is a collection of continuous functions $\phi_1, \dots, \phi_k : X \rightarrow [0, 1]$ s.t.

- $\text{supp}(\phi_i) \subset U_i$
- for all $x \in X$, $\sum_{i=1}^k \phi_i(x) = 1$

Lemma 12.14

Let $\{U_i\}_{i=1}^k$ be a finite covering of X . There exists another open covering $\{V_i\}_{i=1}^k$ s.t. $\overline{V_i} \subset U_i$ for all i

Proof.

Consider the set $X \setminus (U_2 \cup \dots \cup U_k)$. This is a closed subset of U_1 . By normality, $\exists V_1$ s.t. this complement is contained in V_1 and $\overline{V_1} \subset U_1$.

Continue inductively.

Assume $V_1, \dots, V_j, U_{j+1}, \dots, U_k$ is an open covering with $\overline{V_i} \subset U_i$ for $i \leq j$.

As before, replace U_{j+1} with smaller open set V_{j+1} so new collection is still a covering, and $\overline{V_{j+1}} \subset U_{j+1}$. ■

Theorem 12.15

Let X be a normal spaces, let $U = \{U_1, \dots, U_k\}$ be a finite open covering of X . There is a partition of unity subordinate to U .

Proof.

Fix finite covering $U = \{U_1, \dots, U_k\}$. Find open covering $V_1, \dots, V_k$ and $W_1, \dots, W_k$ s.t. $\overline{W_i} \subset V_i$, $\overline{V_i} \subset U_i$.

By Urysohn lemma, there is a function $i : X \rightarrow [0, 1]$ s.t. $f_i(x) > 0$ on W_i , $f_i(x) = 0$ on $X \setminus V_i$.

Thus, $\text{supp}(f_i) \subset U_i$.

Thus, $\sum_{i=1}^k f_i(x) > 0$.

Define: $\phi_i : X \rightarrow [0, 1]$ s.t. $\phi_i(x) = \frac{f_i(x)}{\sum_{i=1}^k f_i(x)}$.

Thus, each ϕ_i is continuous, and the sum is 1 for all $x \in X$. ■

Theorem 12.16

Let M be a compact n -manifold. Then there exists some $N \in \mathbb{N}$ s.t. M embeds into $\mathbb{R}^N$.

Proof.

Let M be a compact n -manifold. Cover M with finitely many open neighborhoods $U_1, \dots, U_k$ that are each homeomorphic to $\mathbb{R}^n$ by $g_i : U_i \rightarrow \mathbb{R}^n$.

Let ϕ_i be a partition of unity subordinate to $\{U_i\}$.

Define functions $h_i : M \rightarrow \mathbb{R}^n$ by: $h_i(x) = \phi_i(x)g_i(x)$ if $x \in U_i$ and 0 if $x \in X \setminus \text{supp}(\phi_i)$.

This is well-defined and continuous.

Define map $F : M \rightarrow \mathbb{R}^k \times \mathbb{R}^{n^k}$ s.t. $F(x) = (\phi_1(x), \dots, \phi_k(x), h_1(x), \dots, h_k(x))$.

This is continuous. Also, injective since if $F(x) = F(y)$ for some $x, y \in M$ then $\phi_i(x) = \phi_i(y)$ for all i , and $\phi_i(x) > 0$ for some i . This implies that $x, y \in U_i$ for some i .

Thus, $h_i(x) = h_i(y) \implies g_i(x) = g_i(y)$ but g_i is a homeomorphism thus $x = y$.

Since M is compact, and $\mathbb{R}^N$ is Hausdorff we are done. Thus, M is an embedding. ■

Theorem 12.17: Tychonoff Theorem

Every product of compact spaces is compact.

13 Spaces of Continuous Functions

Let X be a topological spaces,
let Y be a metric spaces.

$$Y^X = \{ \text{set of functions } X \rightarrow Y \}$$

$$\mathcal{C}(X, Y) = \{f \in Y^X \mid f \text{ is continuous}\}$$

Product topology on Y^X is also called:

- topology of pointwise convergence
- point-open topology

Subbasis for this topology on Y^X :

all sets of the form $S(X, U) = \{f : X \rightarrow Y \mid f(x) \in U\}$
 $x \in X$, and U is open in Y .

This is the same subbasis we used when defining product topology.

Theorem 13.1

Let $f_n : X \rightarrow Y$ be a sequence of functions. Then f_n converges to f in the topology of pointwise convergence if and only if $f_n(x)$ converges to $f(x)$ for all $x \in X$.

Proof.

Follows from the following result:

Let $X = \prod_{\alpha \in J} X_\alpha$, let x_n be a sequence in X , then $x_n \rightarrow x \iff \pi_\alpha(x_n) \rightarrow \pi_\alpha(x)$ for all $\alpha \in J$.

($\implies$) Is immediate since π_α 's are continuous.

($\impliedby$) Let $U = \prod_{\alpha \in J} U_\alpha$ be a basis element containing x .

For each $\alpha \in J$, $\pi_\alpha(x_\alpha) \rightarrow \pi_\alpha(x)$, so there exists N_α s.t. $\pi_\alpha(x_n) \in U_\alpha$ for all $n \geq N_\alpha$.
 $U_\alpha = X_\alpha$ for all but finitely many α , so can take $N_\alpha = 0$ for all but finitely many α .

Thus, $\max_{\alpha \in J} N_\alpha$ exists, call it N .

Thus, $x_n \in U$ for all $n > N$. ■

Consider subset $\mathcal{C}(X, Y) \subset Y^X$. We showed on HW that there is a sequence of continuous functions $f_n : [0, 1] \rightarrow \mathbb{R}$ s.t. f_n converges to $f : [0, 1] \rightarrow \mathbb{R}$, but f is not continuous.

Thus, $\mathcal{C}(X, Y)$ is not closed in Y^X , when Y^X has point-open topology.

Uniform metric topology is called the topology of uniform convergence.

Thus, $\mathcal{C}(X, Y)$ is closed in Y^X .

Proposition 13.2

Let f_n be a sequence of functions then $f_n \rightarrow f$ uniformly $\iff f_n$ converges to f in Y^X with topology of uniform convergence.

By definition, uniform topology on Y^X is metrizable, so it's 1st countable.

Definition 13.3: Topology of compact convergence

Let Y be a metric space, let X be a space. For each $\epsilon > 0$, each compact subset $C \subseteq X$, and each function $f : X \rightarrow Y$, define: $B_C(f, \epsilon) = \text{set of all } g : X \rightarrow Y \text{ s.t. } \sup_{x \in C} d(f(x), g(x)) < \epsilon$.

Proposition 13.4

Let $f_n : X \rightarrow Y$ be a sequence of functions. Then f_n converges to a function $f : X \rightarrow Y$ $\iff$ for every compact subset $C \subset X$, restrictions $f_n|_C$ converge uniformly to $f|_C$.

Definition 13.5: Compactly Generated Spaces

A space X is compactly generated if it satisfies the following:
a set U is open $\iff U \cap C$ is open in C for every compact space $C \subset X$.

Lemma 13.6

Let X, Y be spaces and suppose X is compactly generated. Then a function $f : X \rightarrow Y$ is continuous $\iff f|_C$ is continuous for every compact subspaces $C \subseteq X$.

Proof.

($\implies$) done

($\impliedby$) Let $V \subset Y$ be open, let $C \subset X$ be compact.

$f^{-1}(V) \cap C = (f|_C)^{-1}(V)$ is open in C . Thus, $f^{-1}(V)$ is open since it is open in every compact subspace. ■

Lemma 13.7

IF X is locally compact, or X is 1st countable, then X is compactly generated.

Proof.

Assume X is locally compact. Let $U \subset X$, and suppose $U \cap C$ is open in C for all compact space $C \subset X$.

Fix $x \in U$. Since X is locally compact there is a neighborhood V of x s.t. $\overline{V} \subset U$ for some compact $\overline{V} \subset X$.

Thus, $U \cap \overline{V}$ is open in $\overline{V}$. Thus, $U \cap V$ open in V . Since V is open, this means $U \cap V$ is a neighborhood of x , contained in U . Thus, U is open.

Assume X is 1st countable. Let $A \subset X$, and suppose $A \cap C$ is closed in C for every compact space $C \subset X$.

Let $x \in \overline{A}$. Since first countable there is a sequence $x_n \in A$ s.t. $x_n \rightarrow x$. Consider the set $\{x_n\}_{n \in \mathbb{N}} \cup \{x\} = C$. This is compact.

Thus, $A \cap C$ is closed in C , so the limit x must lie in $A \cap C$. Thus, $x \in A$. So $\overline{A} = A$. Thus, A is closed. ■

Theorem 13.8

If X is compactly generated, Y is metrizable, then (X, Y) is closed in Y^X , if Y^X has topology of compact convergence.

Proof.

Suppose F is in the closure of $\mathcal{C}(X, Y)$. Fix a compact $C \subset X$.

Thus, $B_C(f, 1/n) \cap \mathcal{C}(X, Y)$ is nonempty.

Let f_n be such a continuous function.

Thus, $f_n|_C \rightarrow f|_C$ uniformly.

Thus, $f|_C$ is continuous for every compact $C \subset X$. Since X is compactly generated, means that f is continuous. ■

Definition 13.9

Let X, Y be topological spaces. For each compact subset $C \subset X$, and every open $U \subset Y$, define $S(C, U)$ as

$$\{f \in \mathcal{C}(X, Y) \mid f(C) \subset U\}$$

This forms a subbasis for a topology. Called compact-open topology.

Theorem 13.10

Let X be a space, let Y be a metric spaces. On $\mathcal{C}(X, Y)$, the compact-open topology agrees with topology of compact convergence.

Proof.

Let $f \in S(C, U)$. Since f is continuous, $f(C) \subset U$ is compact.

Then the function $f(C) \rightarrow \mathbb{R}_{\geq 0}$ given by $y \mapsto d(y, X \setminus U)$ has a minimum positive value, call it ϵ .

Thus, $B(f(C), \epsilon) \subset U$. Thus, it follows that $B_C(f, \epsilon) \subset S(C, U)$.

Fix $B_C(f, \epsilon)$. For each point $x \in X$ let V_x be a neighborhood of x s.t. $f(\overline{V_x})$ is contained in an open set U_x with $\text{diam}(U_x) < \epsilon$.

Cover C with finitely many neighborhoods $V_{x_1}, \dots, V_{x_n}$.

Define $C_i = \overline{V_{x_i}} \cap C$. C_i is compact.

If $g \in S(C_i, U_{x_i})$, then for all $x \in C_i$, $d(g(x), f(x)) < \epsilon$.

Thus, $f_1 S(C_1, U_{x_1}) \cap \dots \cap S(C_n, U_{x_n}) \subset B_C(f, \epsilon)$ ■

Corollary 13.11

On $\mathcal{C}(X, Y)$, the topology of compact convergence does not depend on metric on Y .

Corollary 13.12

If X compact, topology of compact convergence is same as topology of uniform convergence $\implies X$ compact $\implies$ compact-open topology is metrizable

Theorem 13.13

Let X be locally compact and Hausdorff.

The evaluation map $e : X \times \mathcal{C}(X, Y) \rightarrow Y$ s.t. $(x, f) \mapsto f(x)$ is continuous, if $\mathcal{C}(X, Y)$ has compact-open topology.

Proof.

Fix $(x, f) \in X \times \mathcal{C}(X, Y)$, V open neighborhood of $f(x)$ in Y .

Pick a neighborhood U of x s.t. $\overline{U}$ is compact and $f(\overline{U}) \subset V$.

$U \times S(\overline{U}, V)$ is an open set containing (x, f) if $y \in U$, $g \in S(\overline{U}, V)$, then $g(y) \in V$ so $(y, g) \in U \times S(\overline{U}, V)$ ■

14 Stone-Čech compactification

Theorem 14.1

There exists a compactification Y of a completely regular X st.t for every $f : X \rightarrow \mathbb{R}$ bounded continuous, there exists $\hat{f} : Y \rightarrow \mathbb{R}$ s.t. $\hat{f}|_X = f$

Proof.

Let $\{f_\alpha\}_{\alpha \in J}$ be all the bounded continuous functions on X . Let $I_\alpha = [\inf f_\alpha(X), \sup f_\alpha(X)]$. Thus, $I = \prod_{\alpha \in J} I_\alpha$ is compact Hausdorff. Define $h : X \rightarrow I$ by $h(x) = (f_\alpha(x))_{\alpha \in J}$. By complete regularity, h is an embedding. Thus, $\overline{h(X)}$ is a compactification. Let $f : X \rightarrow \mathbb{R}$ be a bounded function. Thus, $\exists \beta \in J$ s.t. $f = f_\beta$. Thus, define $\hat{f} : \overline{h(X)} \rightarrow I_\beta$ s.t. $\hat{f}(x) = f(\pi_\beta(x))$. Called $\beta(X)$, Stone-Čech compactification. ■

Theorem 14.2

Suppose $f : X \rightarrow C$, C compact Hausdorff then there exists $\hat{f}$ s.t. $\hat{f} : \beta(X) \rightarrow C$ is continuous and extends f .

Proof.

Since X is completely regular we know that C embeds into some product of $[0,1]$. So assume that $C \subset [0,1]^Y$. Then $f : X \rightarrow C$ can be written as $f_y : X \rightarrow I$, $y \in Y$. Then by the proof above there is a function $\hat{f}_y : \beta(X) \rightarrow I$. Thus, let $\hat{f} = (\hat{f}_y)_{y \in Y}$ ■

Lemma 14.3

Let X be completely regular, and Y be Hausdorff. Let $A \subset X$, $f : A \rightarrow Y$ continuous. Suppose f extends $g : \overline{A} \rightarrow Y$. Then g is unique.

Proof.

Suppose g_1, g_2 extend f . Since they will agree on A suppose $g_1 \neq g_2$ for $x \in \overline{A} \setminus A$. Thus, $g_1(x) \neq g_2(x)$. Thus, we can separate these two points U_1, U_2 open.

Thus, consider $V = g_1^{-1}(U_1) \cap g_2^{-1}(U_2)$. But on $V \cap A$, $g_1 = g_2$. This is a contradiction since $\square$

Theorem 14.4

Suppose Y is a compactification of X . Then, there exists a unique quotient map from $\beta(X) \rightarrow Y$.

15 Algebraic Topology

Definition 15.1: Homotopy

Let X, Y be spaces, let $f_0, f_1 : X \rightarrow Y$ be continuous functions. A homotopy from f_0 to f_1 is a continuous map $X \times [0, 1] \rightarrow Y$ s.t. $h(x, 0) = f_0(x)$ and $h(x, 1) = f_1(x)$.

If X is locally compact and Hausdorff then any continuous path in $\mathcal{C}(X, Y)$ specifies a homotopy between its endpoints.

Definition 15.2: Path Homotopic

Let X be a space, let f_0, f_1 be paths $[0, 1] \rightarrow X$.

We say f_0 and f_1 are path homotopic if they have the same starting point and ending point and there is a homotopy from f_0 to f_1

$h : [0, 1] \times [0, 1] \rightarrow X$ s.t. $h(0, t) = f_0(0) = f_1(0), h(1, t) = f_0(1) = f_1(1)$.

Proposition 15.3

1. Homotopy of maps $X \rightarrow Y$ is an equivalence relation on $\mathcal{C}(X, Y)$
2. Path Homotopy is an equivalence relation on paths $[0, 1] \rightarrow X$.

Proof.

Reflexivity $h : X \times [0, 1] \rightarrow Y$ s.t. $(x, t) \mapsto f(x)$. is a homotopy from f to f .

Symmetry: If f_0, f_1 are homotopic by $h : X \times [0, 1] \rightarrow Y$ then $h' : X \times [0, 1]$ s.t. $h'(x, t) = h(x, 1 - t)$ is a homotopy from f_1 to f_0 .

Reflexivity: If f_0, f_1 are homotopic by $h : X \times [0, 1] \rightarrow Y$ and f_1, f_2 are homotopic by $g : X \times [0, 1] \rightarrow Y$ then $h' : X \times [0, 1]$ s.t. $h'(x, t) = h(x, 2t)$ for $t \leq 1/2$ and $g(x, 2t - 1)$ otherwise. ■

Remark.

In 1, if X is locally compacy Hausdorff, then equivalence classes are the same as path components of $\mathcal{C}(X, Y)$

Example

Fix points $x_0, x_1 \in \mathbb{R}^n$. Any two points between x_0, x_1 are path homotopic. ■

Definition 15.4: Simply Connected

A path-connected spaces is simply connected if any two paths with same endpoints are path-homotopic.

Definition 15.5: Nullhomotopic

A map $f : X \rightarrow Y$ is nullhomotopic if it is homotopic to a constant map.

Definition 15.6: Loop

A loop $f : [0, 1] \rightarrow X$ ($f(0) = f(1)$) is nullhomotopic if it is path-homotopic to a constant map.

15.1 Algebra on Paths and the Fundamental Group**Definition 15.7**

Let X be a space, let f be a path in X from x_0 to x_1 , let g be a path from x_1 to x_2 . Define $f * g$ by.

$$(f * g)(t) = \begin{cases} f(2t) & t \in [0, 1/2] \\ g(2t - 1) & t \in [1/2, 1] \end{cases}$$

Define $*$ on path homotopy classes by:

$$[f] * [g] = [f * g]$$

Warning: We cannot $*$ arbitrary pairs of paths (or arbitrary pairs of path homotopy classes).

Theorem 15.8

On path homotopy classes, the operation $*$ satisfies:

Associativity:

$$([f] * [g]) * [h] = [f] * ([g] * [h])$$

Existence of identities:

Let e_x be the constant path $[0, 1] \rightarrow X, e_x(t) = x$

If f is a path from x_0 to x_1 , then

$$[f] * [e_{x_1}] = [f] = [e_{x_0}] * [f]$$

Inverses:

If f is a path from x_0 to x_1 and $\bar{f}$ is the reverse path then

$$[f] * [\bar{f}] = [e_{x_0}]$$

$$[\bar{f}] * [f] = [e_{x_1}]$$

Definition 15.9: The Fundamental Group

The fundamental group of X relative to basepoint x_0 is the set of path homotopy classes of loops based at x_0 , equipped with group operation $*$.

Denoted $\pi_1(X, x_0)$

Proposition 15.10

If x_0 and x_1 are connected by a path in X , then $\pi_1(X, x_0) \cong \pi_1(X, x_1)$.

Proof.

Let α be a path in X from x_0 to x_1 . Then $\hat{\alpha} : \pi_1(X, x_0) \rightarrow \pi_1(X, x_1)$ s.t. $[f] \mapsto [\bar{\alpha}] * [f] * [\alpha]$.

This is well-define on path-homotopy classes, and it's a group homomorphism:

$[f], [g] \in \pi_1(X, x_0)$ then

$$\hat{\alpha}([f] * [g]) = [\bar{\alpha}] * [f] * [g] * [\alpha] = [\bar{\alpha}] * [f] * [\alpha] * [\bar{\alpha}] * [g] * [\alpha] = \hat{\alpha}([f]) * \hat{\alpha}([g]).$$

Inverse is $[f] \mapsto [\alpha] * [f] * [\bar{\alpha}]$

Definition 15.11

Let $h : X \rightarrow Y$ be a continuous map $h(x_0) = y_0$. The map

$$h_* : \pi_1(X, x_0) \rightarrow \pi_1(Y, y_0)$$

defined by $h_*([f]) = [h \circ f]$

This is a homomorphism.

Theorem 15.12

For continuous maps

$$X \rightarrow Y \rightarrow Z$$

Compositions $k_* \circ h_* = (k \circ h)_*$

Corollary 15.13

If $h : X \rightarrow Y$ is a homeomorphism, $h(x_0) = y_0$, then h_* is an isomorphism.

15.2 Covering Spaces**Definition 15.14: Evenly Covered**

Let X be a topological space. Let $p : \tilde{X} \rightarrow X$ be a surjective continuous map. An open set $U \subset X$ is evenly covered by p if $p^{-1}(U)$ is a disjoint union of open sets $\bigcup_{\alpha \in J} V_\alpha$ s.t. each V_α is homeomorphic to U by p .

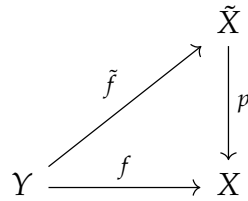
Definition 15.15: Covering Map/Space

p is a covering map if every $x \in X$ has a neighborhood which is evenly covered by p . In this case, $\tilde{X}$ is a covering space of X .

15.3 Lifting Maps**Definition 15.16**

Let $p : \tilde{X} \rightarrow X$ be a continuous map.

Let $f : Y \rightarrow X$. A lift of f is a map $\tilde{f} : Y \rightarrow \tilde{X}$ s.t. $f = p \circ \tilde{f}$.



Theorem 15.17: The Path Lifting Property

Let $p : \tilde{X} \rightarrow X$ be a covering map, let $f : [0, 1] \rightarrow X$ be a path based on $x_0 \in X$. For any point $\tilde{x}_0 \in p^{-1}(x_0)$, there exists a unique path $\tilde{f} : [0, 1] \rightarrow \tilde{X}$ lifting f , based at $\tilde{x}_0$

Proof.

For each point $t \in [0, 1]$ find a neighborhood U_t of $f(t) \in X$ s.t. U_t is evenly covered by p .

The collection of sets $f^{-1}(U_t)$ is an open covering of $[0, 1]$. $[0, 1]$ is compact, so take a finite subcover.

Use this finite subcover to find a sequence of increasing times $0 = t_0 < t_1 < \dots < t_n = 1$ s.t. $[t_i, t_{i+1}]$ satisfied $f([t_i, t_{i+1}])$ lies in a neighborhood U which is evenly covered by p . To construct the lift just lift each subsegment one at a time. ■

Theorem 15.18: The Homotopy Lifting Property

Let $f_0 : X \rightarrow Y$ be a continuous map. Let $p : \tilde{Y} \rightarrow Y$ be a covering map. Suppose $h : X \times [0, 1] \rightarrow Y$ is a homotopy from f_0 to $f_1 : X \rightarrow Y$, and $\tilde{f}_0 : X \rightarrow \tilde{Y}$ is a lift of f_0 . Then there is a unique lift $\tilde{h} : X \times [0, 1] \rightarrow \tilde{Y}$ of h s.t. $\tilde{h}(x, 0) = \tilde{f}_0(x)$

Proof.

(Assume X is locally connected.)

Fix $x_0 \in X$.

Claim

There exists a neighborhood N of x_0 in X s.t. homotopy $h : X \times [0, 1] \rightarrow Y$ "lifts" to a homotopy $\tilde{h} : N \times [0, 1] \rightarrow \tilde{Y}$ s.t. $\tilde{h}(x_0, 0) = \tilde{f}_0(x_0)$.

Proof.

For each $t \in [0, 1]$, find a neighborhood $N_t \subset X$ of x_0 and interval $(t - \epsilon, t + \epsilon)$ s.t. h maps $N_t \times (t - \epsilon, t + \epsilon)$ into a neighborhood of $h(x_0, t)$ in Y , evenly covered by p . This comes from the fact that h is continuous.

$[0, 1]$ is compacy, so we can find a single neighborhood N of x_0 , and times $0 = t_0, \dots, t_n = 1$ s.t. for all i , $h(N \times [t_i, t_{i+1}])$ lies in an evenly covered neighborhood of Y .

WLOG let N be connected. Constrict homotopy in claim one step at a time.

Assume $\tilde{h}$ is defined on $N \times [0, t_i]$ so $\tilde{h}$ is defined on $N \times \{t_i\}$. This is a lift of h on $N \times \{t_i\}$, and $h(N \times \{t_i\})$ lands in an evenly covered neighborhood U_i in Y .

Also, $h(N \times [t_i, t_{i+1}])$ lands in this evenly covered neighborhood. Since N is connected, $\tilde{h}(N \times \{t_i\})$ lands in a unique neighborhood $\tilde{U}_i$ in $\tilde{Y}$ mapping homeomorphically to U_i .

Can compose h with local inverse of covering map on U_i to get a lift of h mapping into $\tilde{U}_i$.

Then by pasting lemma, this gives desired homotopy $\tilde{h} : N \times [0, 1] \rightarrow \tilde{Y}$. ■

From the claim: Define the homotopy $\tilde{h} : X \times [0, 1] \rightarrow \tilde{Y}$ by defining on each neighborhood N of each point $x_0 \in X$.

If we fix $z_0 \in N \cap N'$, then the homotopy from N , and the homotopy from N' define paths in $\tilde{Y}$ based at $\tilde{f}_0(z_0)$, since path lifts a unique these will be the same.

Thus, $\tilde{h} : X \times [0, 1] \rightarrow \tilde{Y}$ is a well-defined, continuous map since it restricts to a continuous map on $N \times [0, 1]$ for neighborhood of points of X . Then by the pasting lemma it is continuous. ■

Let $p : \tilde{X} \rightarrow X$. Let $\tilde{X}$ be path-connected. Fix a basepoint $\tilde{x}_0 \in \tilde{X}$, let $p(\tilde{x}_0) = x_0$.

Given a loop f in X based on x_0 , by the path lifting property we can lift to get a path in $\tilde{X}$ based at $\tilde{x}_0$.

Lifting a loop does not necessarily get a loop, but the endpoint lies in $p^{-1}(x_0)$, so have a map from $\{\text{loops based at } x_0\}$ to $\{\text{points in preimage of } x_0\}$.

This map is well-defined on $\pi_1(X, x_0)$.

Proposition 15.19

There is a well-defined surjective map

$$\phi : \pi_1(X, x_0) \rightarrow p^{-1}(x_0)$$

If $\tilde{X}$ is simply connected, this is a bijection.

Theorem 15.20

$$\pi_1(S^1) \cong (\mathbb{Z}, +)$$

Proof.

Let $p : \mathbb{R} \rightarrow S^1$ s.t. $p(t) = (\cos(2\pi t), \sin(2\pi t))$.

Let $x_0 = p(0)$, and $p^{-1}(x_0) = \mathbb{Z}$. Then we have a bijection:

$$\phi : \pi_1(S^1, x_0) \rightarrow \mathbb{Z}$$

Let $[f], [g] \in \pi_1(S^1, x_0)$. Take unique lifts $\tilde{f}, \tilde{g} : [0, 1] \rightarrow \mathbb{R}$ based at 0.

By definition $\phi([f]) = \tilde{f}(1)$, and $\phi([g]) = \tilde{g}(1)$.

$\tilde{f}(1) \in \mathbb{Z}$ Can define another path $\tilde{\tilde{g}} : [0, 1] \rightarrow \mathbb{R}$ s.t. $\tilde{\tilde{g}}(t) = \tilde{g}(t) + \tilde{f}(1)$. This is a path starting at $\tilde{f}(1)$. So, $\tilde{f} * \tilde{\tilde{g}}$ is well-defined. This is a path based at 0, with an endpoint an integer. So it descends to a loop in S^1 . This loop is homotopic to $f * g$.

$$\phi([f] * [g]) = \phi([f * g]) = (\text{endpoint of lift of } f * g) = \tilde{\tilde{g}}(1) + \tilde{f}(1) + \tilde{g}(1) = \phi([f]) + \phi([g])$$

Given a group G , there is a space X satisfying $\pi_1(X) \cong G$.

Definition 15.21: Retraction

Let X be a space, and $A \subseteq X$. A retraction is a continuous map $r : X \rightarrow A$ s.t. $r|_A$ is identity map on A .

Definition 15.22: Retract

A is a retract of X if there is a retraction $X \rightarrow A$.

Theorem 15.23

Let $r : X \rightarrow A$ be a retraction, let $i : A \rightarrow X$ be inclusion map. Then:

- (i) $i_* : \pi_1(A, a_0) \rightarrow \pi_1(X, a_0)$ is injective.
- (ii) $r_* : \pi_1(X, a_0) \rightarrow \pi_1(A, a_0)$ is surjective.

Proof.

$$id = (r \circ i)_* = r_* \circ i_* \text{ so } i_* \text{ is injective, } r_* \text{ is surjective.}$$

Theorem 15.24

$\mathbb{R}^2 \setminus \{0\}$ is not simply connected.

Proof.

Consider the retraction $r : \mathbb{R}^2 \setminus 0 \rightarrow S^1$ s.t. $r(z) = z/||z||$.

Thus, there is an injective map $\pi_1(S^1) \rightarrow \pi_1(\mathbb{R}^2 \setminus \{0\})$. So $\pi_1(\mathbb{R}^2 \setminus \{0\})$ is nontrivial. ■

Lemma 15.25

No retraction from closed unit disk to S^1

Proof.

If there is then means there is a surjective map from $\pi_1(D) \rightarrow \pi_1(S^1)$ but $\pi_1(D)$ is trivial. ■

Theorem 15.26: Brouwer fixed-point theorem

Any continuous map $f : D \rightarrow D$ has a fixed point.

Proof.

Assume $f : D \rightarrow D$ has no fixed point. Define a retraction $D \rightarrow S^1$: Draw a line between $f(x)$ and x and define $r(x)$ to be the point closer to x . This is a contradiction since there is no retraction. ■

If X is any space homeomorphic to D then any continuous map $X \rightarrow X$ has a fixed point.